

Development of Flow at the Entrance of a Cylindrical Pipe

Numerical Simulation and Modelling of Incompressible Flow

LV-Nr.: 321.053 EX
Winter Semester 2025/26

Felix Scope
`felix_scope@yahoo.de`

September 5, 2026

Contents

Assignment	3
1 Problem Description	6
1.1 Geometry and Setup	6
1.2 Physical Parameters	6
2 Mathematical Formulation	7
2.1 Governing Equations in Cylindrical Coordinates	7
2.1.1 Continuity Equation	7
2.1.2 Momentum Equations	7
2.1.3 Energy Equation	7
2.2 Boundary Conditions	7
2.3 Non-Dimensionalization	8
2.3.1 Reference Scales	8
2.3.2 Non-Dimensional Variables	8
2.3.3 Non-Dimensional Governing Equations	8
2.3.4 Non-Dimensional Boundary Conditions	8
3 Numerical Solution	9
3.1 Computational Domain and Grid	9
3.2 Spatial Discretization: Finite Volume Method	9
3.2.1 Continuity (at scalar CV)	10
3.2.2 Convective Scheme: 2nd-Order Upwind (Linear Upwind Differencing)	10
3.2.3 Axial Momentum (at w-face CV)	10
3.3 Temporal Discretization	11
3.4 Pressure Correction Method (Projection)	11
3.5 Poisson Solver	12
3.6 Ghost Cells and Boundary Implementation	12
4 Implementation	13
4.1 Code Structure	13
4.2 Algorithm Flowchart	15
4.3 Key Implementation Details	15
5 Analytical Solution	17
5.1 Velocity Profile (Hagen–Poiseuille)	17
5.2 Temperature Profile	17
5.3 Validity of the Fully-Developed Temperature Profile	17
5.4 Nusselt Number	18
6 Results	20
6.1 Simulation Parameters	20
6.2 Convergence History	20
6.3 Velocity Field	21
6.4 Temperature Field	22
6.5 Velocity Profiles at Selected Axial Positions	23
6.6 Temperature Profiles at Selected Axial Positions	24

6.7	Comparison with Analytical Solution	25
7	Conclusion	26
A	2D Incompressible Flow in Cylindrical Coordinates (r, z) — Velocities (u, w)	27
A.1	Governing Equations (Strong Form)	27
A.2	Staggered Grid Layout	28
A.3	Finite Volume Discretisation — Integration over $\Delta r \Delta z$	28
A.3.1	Continuity Equation	28
A.3.2	Axial Momentum: CV at $(i, j + \frac{1}{2})$	28
A.3.3	Radial Momentum: CV at $(i + \frac{1}{2}, j)$	29
A.4	Projection Method	29
A.4.1	Step 1 — Predictor	29
A.4.2	Step 2 — Poisson Equation for Pressure Correction p'	30
A.4.3	Step 3 — Velocity Correction	30
A.4.4	Step 4 — Pressure Update	30
A.5	Poisson Solver Procedure	30
A.6	Ghost Cells and Boundary Conditions	31
B	MATLAB Source Code	32
B.1	Main Simulation Script	32
B.2	Grid Generation	36
B.3	Boundary Conditions	36
B.4	Momentum Solver	37
B.5	Poisson Solver and Pressure Correction	42
B.6	Temperature Solver	46
B.7	Convergence Monitoring	48

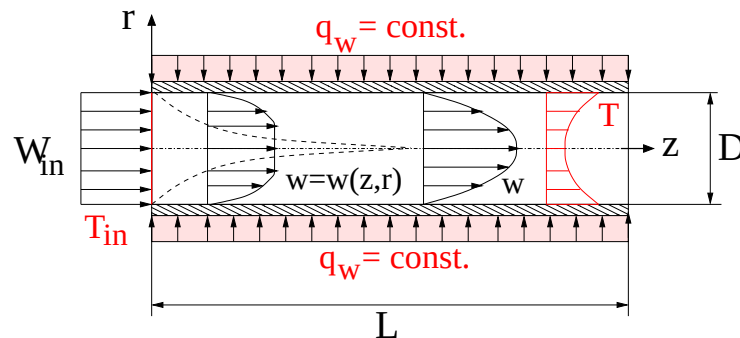
Assignment

Numerical Simulation and Modelling of Incompressible Flow

LV-Nr.: 321.053 **EX**, WS 2025/26

Entwicklung der Strömung am Eintritt eines zylindrischen Rohrs

Es soll eine zwei-dimensionale laminare Strömung in einem zylindrischen Rohr mit gegebenem Durchmesser D simuliert werden. Die gesamte Rohrwand wird mit konstanter Wärmestromdichte $q_w = \text{const.}$ beheizt. Das inkompressible Fluid habe gegebene konstante Dichte $\rho = \text{const.}$, dynamische Viskosität $\mu = \text{const.}$, Wärmeleitfähigkeit $\lambda = \text{const.}$, und spezifische Wärmekapazität $c_v = \text{const.}$. Viskose Dissipation kann in der thermischen Energiebilanz vernachlässigt werden. Die axiale Länge des Rohrs sei $L = 50D$. Die Strömung tritt mit einem gleichförmigen Geschwindigkeits- und Temperaturprofil $w|_{z=0} = W_{in} = \text{const.}$ bzw. $T|_{z=0} = T_{in} = \text{const.}$ in das Rohr ein. Das Strömungsgebiet ist unten stehend schematisch skizziert.



Aufgaben:

- Formulieren Sie die Erhaltungsgleichungen, welche dieses zwei-dimensionale Strömungsproblem beschreiben, in Zylinderkoordinaten.
- Entdimensionalisieren Sie die Bewegungsgleichungen unter Verwendung von D , W_{in} , ρW_{in}^2 , und $q_w / \rho c_v W_{in}$ als Bezugsgrößen für Länge, Geschwindigkeit, Druck, und Temperaturdifferenz $T - T_{in}$.
- Diskretisieren Sie die dimensionslosen Gleichungen
 - räumlich mittels Finite-Volumen Methode zweiter Ordnung genau
 - zeitlich mit einem expliziten Verfahren zweiter Ordnung genau.
- Programmieren Sie ein geeignetes direktes Lösungsverfahren zur Lösung der Poisson Gleichung, wie im Rahmen der Druckkorrektur zur Erfüllung der Kontinuitätsgleichung erforderlich.
- Simulieren Sie die Strömung für den Fall mit $Re = \rho W_{in} D / \mu = 100$, $Pr = \mu c_v / \lambda = 5$ bis das Strömungsfeld einen konvergierten Zustand erreicht hat.
- Dokumentieren Sie Ihre Arbeit. Bei der Darstellung der Ergebnisse sollen insbesondere gezeigt werden:
 - Kontouren der dimensionslosen Geschwindigkeitsfelder in beide Richtungen, also von w/W_{in} (axial) und u/W_{in} (radial), sowie Temperaturfelder $(T(x, y) - T_{in}) \rho W_{in} c_v / q_w$.
 - Profile von $w(r, z) / W_{in}$ und $(T(x, y) - T_{in}) \rho W_{in} c_v / q_w$ an ausgewählten Stromab-Positionen $z_i = \text{const.}$

Korrigierte Seite 2. Seite 1 ist unverändert. Auf dieser Seite ist *nur* die voll entwickelte Vergleichs-Temperaturformel berichtigt: in der Originalfassung fehlte der Faktor $\frac{1}{2}$ vor $Re Pr$ in den radialen Termen, und die additive Konstante muss $-\frac{7}{48}$ statt $-\frac{3}{4}$ lauten, damit die Mischtemperatur $\theta_{\text{bulk}}(z) = 4z/D$ für alle $z \geq 0$ korrekt ist. Die unveränderte Originalfassung liegt als `Assignment_WS25_error.pdf` bei; Herleitung und Literaturquellen ($Nu = 48/11$) in `Report/nusselt_number_analysis.md`.

- Fügen Sie zum Vergleich die Profile für die voll entwickelte zylindrische Rohrströmung hinzu, hier gegeben als

$$w = 2 W_{\text{in}} \left[1 - \left(\frac{2r}{D} \right)^2 \right],$$

$$\frac{(T - T_{\text{in}}) \varrho W_{\text{in}} c_v}{q_w} = 4 \frac{z}{D} + Re Pr \left[\frac{1}{2} \left(\frac{2r}{D} \right)^2 - \frac{1}{8} \left(\frac{2r}{D} \right)^4 - \frac{7}{48} \right],$$

welche für zunehmende Lauflänge z schlussendlich erreicht werden müssen.

Äquivalente Schreibweise (Faktor ausgeklammert):

$$\frac{(T - T_{\text{in}}) \varrho W_{\text{in}} c_v}{q_w} = 4 \frac{z}{D} + \frac{1}{2} Re Pr \left[\left(\frac{2r}{D} \right)^2 - \frac{1}{4} \left(\frac{2r}{D} \right)^4 - \frac{7}{24} \right].$$

1 Problem Description

1.1 Geometry and Setup

The problem considers two-dimensional laminar flow in a cylindrical pipe with diameter D and length $L = 50D$. The pipe wall is heated with constant heat flux $q_w = \text{const.}$. The incompressible fluid has constant properties: density ρ , dynamic viscosity μ , thermal conductivity λ , and specific heat capacity c_v . Viscous dissipation is neglected.

The flow enters with uniform velocity and temperature profiles:

$$w|_{z=0} = W_{\text{in}} = \text{const.} \quad (1)$$

$$T|_{z=0} = T_{\text{in}} = \text{const.} \quad (2)$$

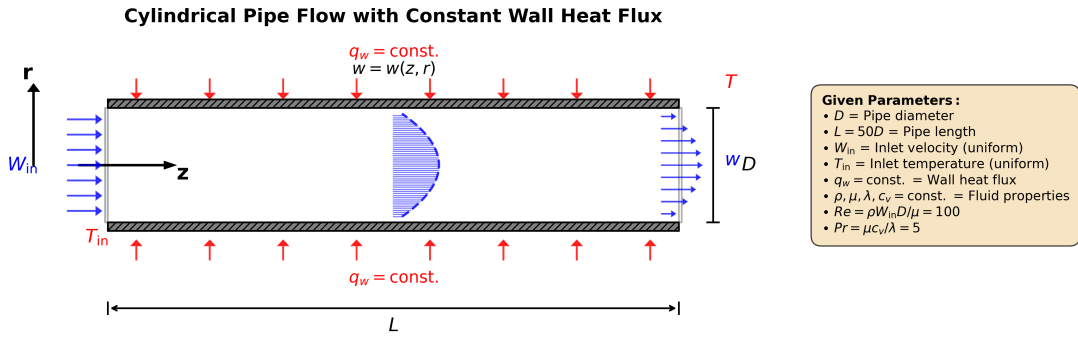


Figure 1: Schematic of the cylindrical pipe flow with constant wall heat flux. Flow enters with uniform velocity W_{in} and temperature T_{in} , and develops into a parabolic profile along the pipe length.

1.2 Physical Parameters

- Reynolds number: $Re = \frac{\rho W_{\text{in}} D}{\mu} = 100$
- Prandtl number: $Pr = \frac{\mu c_v}{\lambda} = 5$
- Length-to-diameter ratio: $L/D = 50$

2 Mathematical Formulation

2.1 Governing Equations in Cylindrical Coordinates

For axisymmetric flow in cylindrical coordinates (r, z) with velocity components (u, w) in the radial and axial directions, the governing equations in **conservation form** are:

2.1.1 Continuity Equation

$$\frac{1}{r} \frac{\partial(r u)}{\partial r} + \frac{\partial w}{\partial z} = 0 \quad (3)$$

2.1.2 Momentum Equations

Radial momentum:

$$\frac{\partial u}{\partial t} + \frac{1}{r} \frac{\partial(r u^2)}{\partial r} + \frac{\partial(u w)}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial r} + \frac{1}{\rho} \left[\frac{1}{r} \frac{\partial(r \tau_{rr})}{\partial r} - \frac{\tau_{\varphi\varphi}}{r} + \frac{\partial \tau_{rz}}{\partial z} \right] \quad (4)$$

Axial momentum:

$$\frac{\partial w}{\partial t} + \frac{1}{r} \frac{\partial(r u w)}{\partial r} + \frac{\partial(w^2)}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial z} + \frac{1}{\rho} \left[\frac{1}{r} \frac{\partial(r \tau_{rz})}{\partial r} + \frac{\partial \tau_{zz}}{\partial z} \right] \quad (5)$$

where the stress tensor components for a Newtonian fluid are:

$$\tau_{rr} = 2\mu \frac{\partial u}{\partial r}, \quad \tau_{\varphi\varphi} = 2\mu \frac{u}{r}, \quad \tau_{zz} = 2\mu \frac{\partial w}{\partial z}, \quad \tau_{rz} = \mu \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial r} \right) \quad (6)$$

2.1.3 Energy Equation

$$\frac{\partial T}{\partial t} + \frac{1}{r} \frac{\partial(r u T)}{\partial r} + \frac{\partial(w T)}{\partial z} = \frac{\lambda}{\rho c_v} \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{\partial^2 T}{\partial z^2} \right] \quad (7)$$

2.2 Boundary Conditions

Inlet ($z = 0$):

$$u = 0, \quad w = W_{\text{in}}, \quad T = T_{\text{in}} \quad (8)$$

Wall ($r = D/2$):

$$u = 0, \quad w = 0, \quad \lambda \frac{\partial T}{\partial r} \Big|_{r=D/2} = q_w \quad (9)$$

Heat flows into the fluid at the wall (constant heating), so with r increasing outward this is a positive gradient: the wall is hotter than the centreline.

Centreline ($r = 0$, symmetry):

$$u = 0, \quad \frac{\partial w}{\partial r} \Big|_{r=0} = 0, \quad \frac{\partial T}{\partial r} \Big|_{r=0} = 0 \quad (10)$$

Outlet ($z = L$, zero-gradient):

$$\frac{\partial u}{\partial z} = 0, \quad \frac{\partial w}{\partial z} = 0, \quad \frac{\partial T}{\partial z} = 0 \quad (11)$$

2.3 Non-Dimensionalization

2.3.1 Reference Scales

- Length: D
- Velocity: W_{in}
- Pressure: ρW_{in}^2
- Temperature difference: $\frac{q_w}{\rho c_v W_{\text{in}}}$
- Time: D/W_{in}

2.3.2 Non-Dimensional Variables

$$r^* = \frac{r}{D}, \quad z^* = \frac{z}{D}, \quad t^* = \frac{t W_{\text{in}}}{D} \quad (12)$$

$$u^* = \frac{u}{W_{\text{in}}}, \quad w^* = \frac{w}{W_{\text{in}}}, \quad p^* = \frac{p}{\rho W_{\text{in}}^2} \quad (13)$$

$$\theta = \frac{(T - T_{\text{in}}) \rho c_v W_{\text{in}}}{q_w} \quad (14)$$

2.3.3 Non-Dimensional Governing Equations

Dropping asterisks for brevity (all quantities are non-dimensional hereafter):

Continuity:

$$\frac{1}{r} \frac{\partial(r u)}{\partial r} + \frac{\partial w}{\partial z} = 0 \quad (15)$$

Radial momentum:

$$\frac{\partial u}{\partial t} + \frac{1}{r} \frac{\partial(r u^2)}{\partial r} + \frac{\partial(u w)}{\partial z} = -\frac{\partial p}{\partial r} + \frac{1}{\text{Re}} \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) - \frac{u}{r^2} + \frac{\partial^2 u}{\partial z^2} \right] \quad (16)$$

Axial momentum:

$$\frac{\partial w}{\partial t} + \frac{1}{r} \frac{\partial(r u w)}{\partial r} + \frac{\partial(w^2)}{\partial z} = -\frac{\partial p}{\partial z} + \frac{1}{\text{Re}} \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial w}{\partial r} \right) + \frac{\partial^2 w}{\partial z^2} \right] \quad (17)$$

Energy:

$$\frac{\partial \theta}{\partial t} + \frac{1}{r} \frac{\partial(r u \theta)}{\partial r} + \frac{\partial(w \theta)}{\partial z} = \frac{1}{\text{RePr}} \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \theta}{\partial r} \right) + \frac{\partial^2 \theta}{\partial z^2} \right] \quad (18)$$

2.3.4 Non-Dimensional Boundary Conditions

Inlet ($z = 0$): $u = 0, w = 1, \theta = 0$

Wall ($r = 1/2$): $u = 0, w = 0, \partial \theta / \partial r = \text{RePr}$

Non-dimensionalizing the wall condition directly: $\partial \theta / \partial r^* = D \cdot (\rho c_v W_{\text{in}} / q_w) \cdot \partial T / \partial r|_{\text{wall}} = D \cdot (\rho c_v W_{\text{in}} / q_w) \cdot (q_w / \lambda) = \rho c_v W_{\text{in}} D / \lambda = \text{RePr}$ (i.e. 500 for $\text{Re} = 100, \text{Pr} = 5$), *not* ± 1 — the reference length/temperature scales do not cancel.

Centreline ($r = 0$): $u = 0, \partial w / \partial r = 0, \partial \theta / \partial r = 0$

Outlet ($z = 50$): $\partial u / \partial z = 0, \partial w / \partial z = 0, \partial \theta / \partial z = 0$

3 Numerical Solution

3.1 Computational Domain and Grid

The computational domain is:

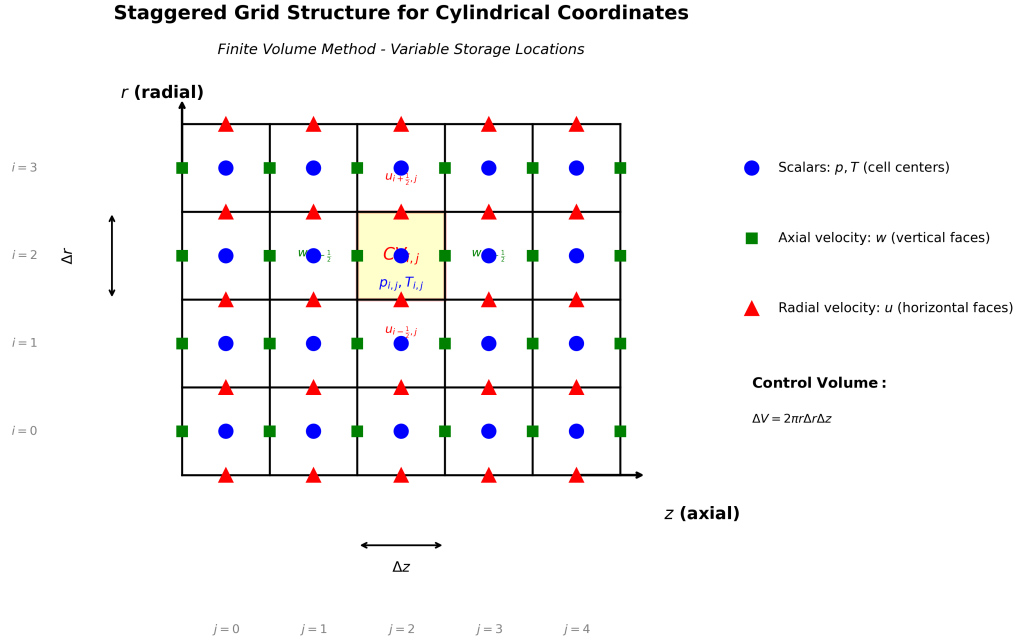
$$\Omega = \{(r, z) : 0 \leq r \leq 0.5, \quad 0 \leq z \leq 50\} \quad (19)$$

A **staggered grid** (MAC grid) is used:

- Scalar variables (p, θ): cell centres ($r_{c,i}, z_{c,j}$)
- Radial velocity u : radial faces ($r_{f,i}, z_{c,j}$)
- Axial velocity w : axial faces ($r_{c,i}, z_{f,j}$)

Grid parameters: $n_r = 50$ cells radially, $n_z = 250$ cells axially, giving:

$$\Delta r = \frac{0.5}{n_r} = 0.01, \quad \Delta z = \frac{50}{n_z} = 0.2 \quad (20)$$



Note : Staggered arrangement prevents pressure-velocity coupling issues
Velocity components stored at control volume faces where fluxes are computed

Figure 2: Staggered grid structure. Scalars (p, θ) at cell centres, axial velocity w at vertical faces, radial velocity u at horizontal faces.

3.2 Spatial Discretization: Finite Volume Method

The governing equations are integrated over each control volume using the FVM. After applying Gauss' divergence theorem, the volume integrals become surface flux integrals that are evaluated at control-volume faces.

3.2.1 Continuity (at scalar CV)

$$\frac{r_{i+\frac{1}{2}} u_{i+1,j} - r_{i-\frac{1}{2}} u_{i,j}}{r_{c,i} \Delta r} + \frac{w_{i,j+1} - w_{i,j}}{\Delta z} = 0 \quad (21)$$

3.2.2 Convective Scheme: 2nd-Order Upwind (Linear Upwind Differencing)

Central differencing requires the cell Peclet number $Pe = |v| \Delta x / (\nu_{\text{eff}})$ to satisfy $Pe < 2$. For our grid:

$$Pe_z = |w| \Delta z Re = 1 \times 0.2 \times 100 = 20 \gg 2 \quad (22)$$

(and $Pe_\theta = |w| \Delta z RePr = 100$ for the energy equation, even higher).

Pure central differencing is therefore prone to non-physical oscillations at these cell Peclet numbers. We use **2nd-order upwind (linear-upwind / LUD) differencing** for convective terms: at each face, the transported quantity is linearly extrapolated from the two cells on the *upwind* side of the face, based on the sign of the transporting velocity. This keeps the upwind stability bias while being formally 2nd-order accurate, matching the assignment's FVM accuracy requirement (an earlier version of this solver used plain 1st-order upwind, which only satisfies this requirement for the diffusive terms).

For a generic flux $F = v \phi$ at the face between cells i and $i + 1$:

$$\phi_{\text{face}} = \begin{cases} \frac{3}{2} \phi_i - \frac{1}{2} \phi_{i-1} & \text{if } v_{\text{face}} > 0 \\ \frac{3}{2} \phi_{i+1} - \frac{1}{2} \phi_{i+2} & \text{if } v_{\text{face}} < 0 \end{cases} \quad (23)$$

which reduces to the classic 1st-order upwind value (ϕ_i or ϕ_{i+1}) if the linear extrapolation term is dropped.

Diffusive terms retain standard **second-order central differencing**.

3.2.3 Axial Momentum (at w-face CV)

The w-CV at $(r_{c,i}, z_{f,j})$ spans $[r_{f,i}, r_{f,i+1}] \times [z_{c,j-1}, z_{c,j}]$.

AB2 predictor step (including the known pressure gradient):

$$w_{i,j}^* = w_{i,j}^n + \Delta t \left[\frac{3}{2} F_{w,i,j}^n - \frac{1}{2} F_{w,i,j}^{n-1} - \frac{p_{i,j}^n - p_{i,j-1}^n}{\Delta z} \right], \quad F_w = -C_r^w - C_z^w + D_r^w + D_z^w \quad (24)$$

where the convective terms (upwind) are:

$$C_r^w = \frac{r_{i+\frac{1}{2}} u_E w_{E,\text{up}} - r_{i-\frac{1}{2}} u_I w_{I,\text{up}}}{r_{c,i} \Delta r} \quad (25)$$

$$C_z^w = \frac{\bar{w}_N w_{N,\text{up}} - \bar{w}_S w_{S,\text{up}}}{\Delta z} \quad (26)$$

and the diffusive terms (central) are:

$$D_r^w = \frac{1}{Re} \frac{r_{i+\frac{1}{2}} (w_{i+1,j} - w_{i,j}) - r_{i-\frac{1}{2}} (w_{i,j} - w_{i-1,j})}{r_{c,i} \Delta r^2} \quad (27)$$

$$D_z^w = \frac{1}{Re} \frac{w_{i,j+1} - 2w_{i,j} + w_{i,j-1}}{\Delta z^2} \quad (28)$$

The radial momentum equation is discretized analogously at u-face control volumes.

3.3 Temporal Discretization

Explicit second-order Adams–Bashforth time stepping is used. For a spatial operator F , the update is

$$F^{AB2} = \frac{3}{2}F^n - \frac{1}{2}F^{n-1}, \quad (29)$$

$$\phi^{n+1} = \phi^n + \Delta t F^{AB2}. \quad (30)$$

The first step uses forward Euler because F^{n-1} is not yet available. The current momentum and temperature RHS arrays are stored for the following step. AB2 needs one RHS evaluation and one pressure-Poisson solve per time step. The temperature equation uses the same AB2 extrapolation of its convection–diffusion operator.

The time step $\Delta t = 0.002$ satisfies the stability constraints:

- Convective CFL: $\Delta t / \Delta z = 0.01 \ll 1$
- Viscous stability: $\Delta t (1/\text{Re}) (1/\Delta r^2 + 1/\Delta z^2) \approx 0.2 < 0.5$

3.4 Pressure Correction Method (Projection)

A segregated approach is used. Each time step consists of:

1. **Predictor:** Extrapolate the convective–diffusive momentum RHS with AB2 and include the known pressure gradient:

$$\vec{u}^* = \vec{u}^n + \Delta t \left[\frac{3}{2}F^n - \frac{1}{2}F^{n-1} - \nabla p^n \right]. \quad (31)$$

2. **Poisson equation:** Find pressure correction p' such that the corrected velocity is divergence-free:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial p'}{\partial r} \right) + \frac{\partial^2 p'}{\partial z^2} = \frac{1}{\Delta t} \left[\frac{1}{r} \frac{\partial (r u^*)}{\partial r} + \frac{\partial w^*}{\partial z} \right] \quad (32)$$

Discretized:

$$\frac{r_{i+\frac{1}{2}}(p'_{i+1,j} - p'_{i,j}) - r_{i-\frac{1}{2}}(p'_{i,j} - p'_{i-1,j})}{r_{c,i} \Delta r^2} + \frac{p'_{i,j+1} - 2p'_{i,j} + p'_{i,j-1}}{\Delta z^2} = b_{i,j} \quad (33)$$

3. **Velocity correction:**

$$u_{i,j}^{n+1} = u_{i,j}^* - \Delta t \frac{p'_{i,j} - p'_{i-1,j}}{\Delta r} \quad (34)$$

$$w_{i,j}^{n+1} = w_{i,j}^* - \Delta t \frac{p'_{i,j} - p'_{i,j-1}}{\Delta z} \quad (35)$$

4. **Pressure update** with under-relaxation:

$$p^{n+1} = p^n + \alpha_p p', \quad \alpha_p = 0.7 \quad (36)$$

3.5 Poisson Solver

The Poisson equation (32) is assembled into a sparse linear system $\mathbf{A} \mathbf{p}' = \mathbf{b}$ of size $(n_r \cdot n_z) \times (n_r \cdot n_z)$. The matrix $\mathbf{A}$ is assembled once and reused every time step. It is solved using MATLAB's built-in **direct sparse solver** (backslash operator $\mathbf{A} \backslash \mathbf{b}$), which employs LU decomposition for the sparse system.

Boundary conditions for p' :

- Centreline ($r = 0$): Neumann $\partial p' / \partial r = 0$ (natural: $r_{-\frac{1}{2}} = 0$, flux vanishes)
- Wall ($r = 0.5$): Neumann $\partial p' / \partial r = 0$
- Inlet ($z = 0$): Neumann $\partial p' / \partial z = 0$
- Outlet ($z = 50$): Dirichlet $p' = 0$ (reference pressure)

3.6 Ghost Cells and Boundary Implementation

Ghost cells are used to impose boundary conditions in the staggered grid. For each boundary, a ghost cell value is defined such that the appropriate condition is satisfied at the boundary face.

Table 1: Ghost cell values for each boundary.

Boundary	u_{ghost}	w_{ghost}	p_{ghost}	θ_{ghost}
Centreline ($r = 0$)	$u_g = 0$	$w_g = w_1$ (symmetry)	$p_g = p_1$	$\theta_g = \theta_1$
Wall ($r = 0.5$)	$u_g = 0$	$w_g = -w_{n_r}$ (no-slip)	$p_g = p_{n_r}$	$\theta_g = \theta_{n_r} + \Delta r \cdot \text{RePr}$
Inlet ($z = 0$)	$u_g = -u_1$	$w = 1$ (Dirichlet)	$p_g = p_1$	$\theta = 0$ (Dirichlet)
Outlet ($z = L$)	$u_g = u_{n_z}$	$w_g = w_{n_z}$	$p' = 0$	$\theta_g = \theta_{n_z}$

The wall temperature ghost cell $\theta_g = \theta_{n_r} + \Delta r \cdot \text{RePr}$ implements the Neumann condition $\partial \theta / \partial r = \text{RePr}$ via central differencing:

$$\left. \frac{\partial \theta}{\partial r} \right|_{\text{wall}} \approx \frac{\theta_g - \theta_{n_r}}{\Delta r} = \frac{\Delta r \cdot \text{RePr}}{\Delta r} = \text{RePr} \quad (37)$$

An earlier version of this derivation incorrectly dropped the RePr factor (giving $\partial \theta / \partial r = -1$); see `Report/temperature_formula_review.md` for the full re-derivation and cross-checks.

4 Implementation

4.1 Code Structure

The solver is implemented in MATLAB. Table 2 lists all functions and their roles.

Table 2: MATLAB function structure.

Function	Description
<code>constants.m</code>	Returns struct with physical parameters (Re, Pr), grid sizes, time step, tolerances
<code>make_grid.m</code>	Generates staggered grid: $r_c, r_f, z_c, z_f, \Delta r, \Delta z$
<code>initialise_fields.m</code>	Allocates fields (u, w, p, θ) with uniform initial conditions
<code>apply_bc_w.m</code>	Enforces axial velocity BCs (inlet Dirichlet, outlet zero-gradient)
<code>apply_bc_u.m</code>	Enforces radial velocity BCs (axis and wall: $u = 0$)
<code>apply_bc_T.m</code>	Enforces temperature BCs (inlet Dirichlet, outlet zero-gradient)
<code>apply_all_bc.m</code>	Dispatches to <code>apply_bc_u/w/T.m</code> in one call (used for the initial-condition BCs before the time loop)
<code>upwind2_face.m</code>	2nd-order upwind (linear-extrapolation) face-value helper, shared by all three RHS functions below
<code>compute_rhs_w.m</code>	Explicit RHS of axial momentum (no pressure): 2nd-order upwind convection, central diffusion
<code>compute_rhs_u.m</code>	Explicit RHS of radial momentum (no pressure): 2nd-order upwind convection, central diffusion
<code>compute_rhs_T.m</code>	Explicit RHS of the energy equation: 2nd-order upwind convection, central diffusion
<code>predictor_w.m</code>	Thin 1-stage-Euler wrapper around <code>compute_rhs_w.m</code> (kept for standalone use)
<code>predictor_u.m</code>	Thin 1-stage-Euler wrapper around <code>compute_rhs_u.m</code> (kept for standalone use)
<code>build_poisson_matrix.m</code>	Assembles sparse Poisson matrix $\mathbf{A}$ (called once)
<code>solve_poisson.m</code>	Solves $\mathbf{A} \mathbf{p}' = \mathbf{b}$ using MATLAB backslash
<code>correct_velocity.m</code>	Applies pressure-gradient correction to u^*, w^*
<code>update_pressure.m</code>	Updates pressure: $p^{n+1} = p^n + \alpha_p p'$
<code>advance_temperature.m</code>	Thin 1-stage-Euler wrapper around <code>compute_rhs_T.m</code> (kept for standalone use)
<code>ab2_step.m</code>	Advances (u, w, p, θ) with AB2 and one incremental pressure correction; stores the current RHS for the next step
<code>compute_divergence.m</code>	Computes discrete divergence of velocity field
<code>compute_residuals.m</code>	Computes convergence residuals ($R_{\text{cont}}, R_{\text{vel}}, R_{\theta}$)
<code>check_convergence.m</code>	Returns true when all residuals are below tolerances
<code>analytical_velocity.m</code>	Hagen–Poiseuille velocity profile
<code>analytical_temperature.m</code>	Analytical temperature profile for constant wall heat flux
<code>main.m</code>	Main script: initialisation $\rightarrow$ time stepping $\rightarrow$ post-processing and plotting

4.2 Algorithm Flowchart

Figure 3 shows the algorithm implemented in `main.m`.

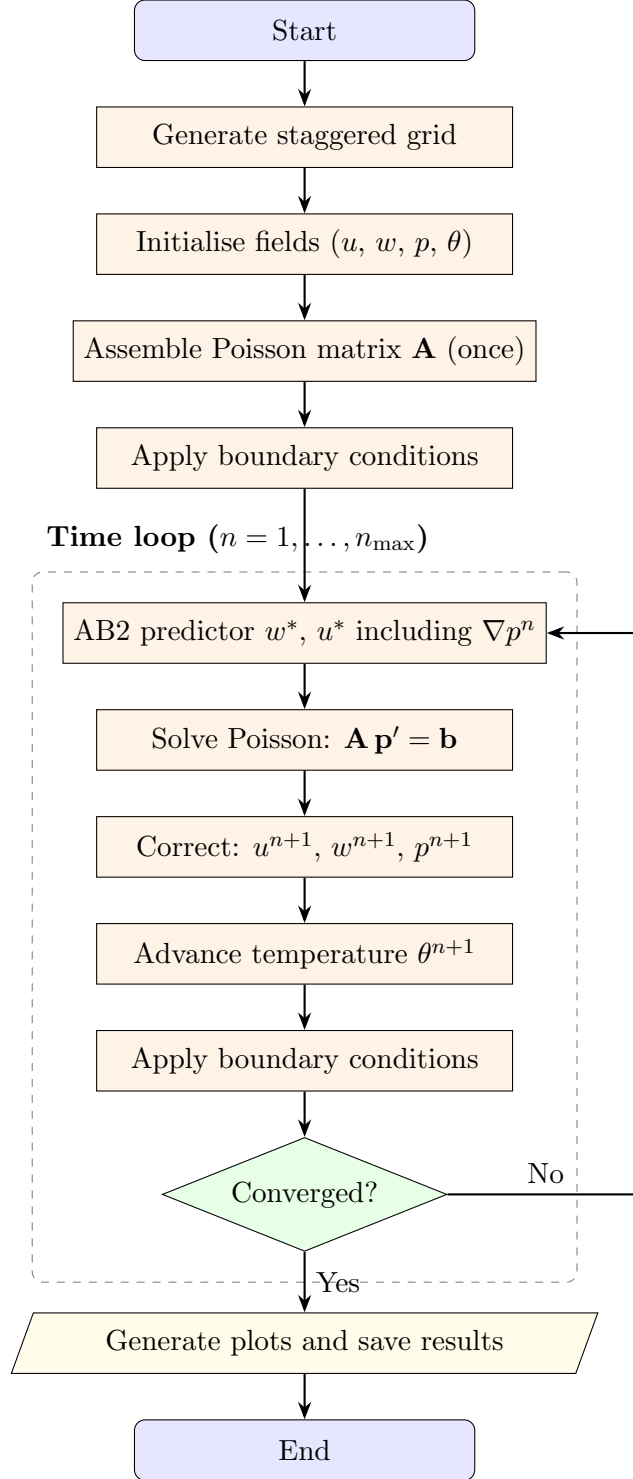


Figure 3: Algorithm flowchart for the pipe flow solver.

4.3 Key Implementation Details

- **2nd-order upwind differencing:** Convective fluxes in both momentum and energy equations use 2nd-order (linear-extrapolation) upwind to avoid oscillations at

high Peclet numbers ($Pe_z = 20$ for momentum, $Pe_\theta = 100$ for energy) while remaining formally 2nd-order accurate.

- **Sparse direct solver:** The Poisson matrix ($12,500 \times 12,500$) is assembled once using MATLAB's `sparse` function and solved each time step with the backslash operator (`A\b`).
- **Vectorised operations:** All stencil computations are performed using MATLAB's array operations, avoiding explicit loops over grid cells for efficiency.
- **Staggered grid:** Prevents checkerboard pressure oscillations; velocity components are stored at the faces where mass fluxes are naturally evaluated.
- **Ghost cells:** All boundary conditions are implemented via ghost-cell extensions. The arrays are extended by one row/column using MATLAB array concatenation, and the boundary condition determines the ghost value. This avoids special-casing inside the stencil computations.

5 Analytical Solution

For fully developed flow (large z), analytical solutions exist for comparison.

5.1 Velocity Profile (Hagen–Poiseuille)

$$\frac{w}{W_{\text{in}}} = 2 [1 - (2r^*)^2] \quad (38)$$

The centreline velocity is $w_{\text{max}} = 2 W_{\text{in}}$. The hydrodynamic entry length is approximately $L_h \approx 0.05 \text{ Re } D = 5D$.

5.2 Temperature Profile

For constant wall heat flux in fully developed thermal flow:

$$\theta = 4z^* + \text{RePr} \left[\frac{1}{2}(2r^*)^2 - \frac{1}{8}(2r^*)^4 - \frac{7}{48} \right] \quad (39)$$

This form follows from direct integration of the non-dimensional energy equation (18) in the fully developed regime (Section 5.4), subject to the wall condition $\lambda \partial T / \partial r|_{\text{wall}} = q_w$ and the global energy balance $\theta_{\text{bulk}}(z^*) = 4z^*$; see `Report/nusselt_number_analysis.md` for the full derivation and literature sources.

The thermal entry length is $L_t \approx 0.05 \text{ Re Pr } D = 25D$.

5.3 Validity of the Fully-Developed Temperature Profile

Equation (39) makes two logically separate claims, which hold over different ranges of z^* because they rest on different physics: one about the cross-sectional *level* of the temperature (the bulk value), and one about its *radial shape*.

The bulk temperature is exact from the inlet. The bulk (mixing-cup) temperature is the flow-weighted cross-sectional average

$$\theta_{\text{bulk}}(z^*) = \frac{\int w^* \theta \, dA}{\int w^* \, dA}. \quad (40)$$

Its axial evolution follows from an energy balance on a thin slice of pipe — enthalpy gained by the stream equals heat entering through the wall:

$$\dot{m} c_v \frac{\partial T_{\text{bulk}}}{\partial z} = q_w \pi D. \quad (41)$$

Every factor on the right is constant (q_w is constant and applied from $z = 0$; the perimeter is fixed), and $\dot{m}$, c_v are constant, so $\partial T_{\text{bulk}} / \partial z$ is constant along the *entire* pipe. Non-dimensionalized this is $\partial \theta_{\text{bulk}} / \partial z^* = 4$, and integrating from $\theta_{\text{bulk}}(0) = 0$ gives

$$\theta_{\text{bulk}}(z^*) = 4z^* \quad \text{exactly, for all } z^* \geq 0, \quad (42)$$

developed or not. This balance involves only totals (total wall heat input, total enthalpy flux); it never refers to the shape of the radial profile, so the thermal entrance region does not perturb it. The one effect that would — axial conduction near the inlet — is neglected here, as specified.

The radial shape requires thermally developed flow. Reducing the energy equation (18) to the closed r^* -form in (39) assumes the profile is *thermally fully developed*: self-similar in shape, merely translating downstream, so $\partial(\theta - \theta_{\text{bulk}})/\partial z^* = 0$. That is exactly the step that makes $\partial\theta/\partial z^*$ the r^* -independent constant 4 and collapses the PDE to an ODE in r^* . Near the inlet it is false: at $z^* = 0$ the temperature is uniform ($\theta = 0$, flat); the developed shape has the wall hotter than the centreline by $\frac{11}{48}\text{RePr}$; and over the thermal entry length $L_t \approx 25D$ the profile morphs continuously between the two, a thin heated layer growing inward from the wall while the core stays at inlet temperature. Through this region $\partial(\theta - \theta_{\text{bulk}})/\partial z^*$ is large and r^* -dependent, and the ODE solution does not apply. Evaluated there anyway, (39) gives at the centreline $4z^* + \text{RePr}(-\frac{7}{48}) \approx 4z^* - 73$, i.e. negative for $z^* \lesssim 18$ — claiming the core is colder than the inlet. The numerical solution instead holds the core near $\theta = 0$ until the thermal layer reaches it ($z^* \approx 25$), then bends toward the analytical curve, the two nearly coinciding by $z^* = 50$ (Fig. 9). By contrast the hydrodynamic entry length is only $L_h \approx 0.05 \text{Re} D \approx 5D$: velocity develops about five times sooner, since here momentum diffuses $\text{Pr} = 5$ times faster than heat.

Why the level can be right while the shape is not. The constant $-\frac{7}{48}$ is fixed by requiring the flow-weighted average of the radial bracket to vanish, so (39) is *built* to satisfy $\theta_{\text{bulk}} = 4z^*$; its mean is pinned to the correct value even where its distribution about that mean is wrong. This mirrors the Hagen–Poiseuille profile (38), which always carries the correct flow rate (mean $w^* = 1$ by construction) although the true profile near the inlet is plug-like, not parabolic — right integral, wrong shape. The fully-developed profile is therefore used only as the $z^* \rightarrow \infty$ reference it is: its departure from the numerical solution at small z^* is expected behaviour, not a defect, and resolving the entrance region is the numerical solution’s role.

5.4 Nusselt Number

For hydrodynamically and thermally fully developed laminar flow in a circular pipe with *constant wall heat flux*, the Nusselt number is the exact rational value

$$\text{Nu} = \frac{hD}{\lambda} = \frac{q_w D}{\lambda(T_{\text{wall}} - T_{\text{bulk}})} = \frac{\text{RePr}}{\theta_{\text{wall}} - \theta_{\text{bulk}}} = \frac{48}{11} \approx 4.3636, \quad (43)$$

a standard textbook result for this thermal boundary condition [1, 2].

Derivation. In the fully developed region $u^* = 0$ and the profile translates rigidly in z^* , so the non-dimensional energy equation (18) reduces to

$$w^*(r^*) \frac{\partial\theta}{\partial z^*} = \frac{1}{\text{RePr}} \frac{1}{r^*} \frac{\partial}{\partial r^*} \left(r^* \frac{\partial\theta}{\partial r^*} \right). \quad (44)$$

A global energy balance over a pipe slice fixes the axial gradient: with the wall adding heat at non-dimensional rate $\partial\theta/\partial r^*|_{\text{wall}} = \text{RePr}$ and the flow carrying mean velocity $\langle w^* \rangle = 1$, the bulk (mixing-cup) temperature satisfies $\theta_{\text{bulk}}(z^*) = 4z^*$ exactly, hence $\partial\theta/\partial z^* = 4$ everywhere in the cross-section. Substituting this and $w^* = 2[1 - (2r^*)^2]$ into (44) and integrating twice, with $\partial\theta/\partial r^*|_{r^*=0} = 0$ and $\partial\theta/\partial r^*|_{r^*=1/2} = \text{RePr}$,

$$\theta(r^*, z^*) = 4z^* + \text{RePr} \left[\frac{1}{2}(2r^*)^2 - \frac{1}{8}(2r^*)^4 + c \right], \quad (45)$$

and the constant c follows from requiring the flow-weighted average of the bracket to vanish (so that $\theta_{\text{bulk}} = 4z^*$), which gives $c = -\frac{7}{48}$ — recovering Eq. (39). Evaluating the bracket at the wall gives $\theta_{\text{wall}} - \theta_{\text{bulk}} = \frac{11}{48} \text{RePr}$, and (43) follows.

Consistency check. The value is exact: (44) with its boundary conditions is a linear problem with no free parameters, so Nu is independent of Re , Pr , q_w , the fluid properties and the grid. Taking instead the assignment sheet’s literal radial coefficients $(2r^*)^2 - \frac{1}{4}(2r^*)^4 - \frac{3}{4}$ yields $\theta_{\text{wall}} - \theta_{\text{bulk}} = \frac{11}{24} \text{RePr}$ and hence $\text{Nu} = 24/11 \approx 2.18$ — exactly half of $48/11$, and inconsistent both with this classical result and with the assignment’s own stated wall condition (that literal profile has wall gradient 2RePr , and its bulk temperature is not $4z^*$). The converged CFD solution gives $\text{Nu} \approx 4.4\text{--}4.6$ over $z^* = 20\text{--}40$, approaching $48/11$ where the thermal field has had the most settling time. The full derivation, the literature sources, and the empirical table are collected in `Report/nusselt_number_analysis.md`.

6 Results

6.1 Simulation Parameters

Table 3: Simulation parameters.

Parameter	Value	Description
Re	100	Reynolds number
Pr	5	Prandtl number
$n_r \times n_z$	50×250	Grid resolution
Δr	0.01	Radial grid spacing
Δz	0.2	Axial grid spacing
Δt	0.002	Time step
n_{steps}	30 000	Maximum time steps
α_p	0.7	Pressure under-relaxation
Convective scheme	2nd-order upwind (LUD)	Stability for $Pe \gg 2$, 2nd-order accurate
Time integration	Explicit Adams–Bashforth 2	2nd-order; Euler startup

6.2 Convergence History

The simulation targets convergence when the continuity, velocity, and temperature residuals fall below 10^{-6} :

$$R_{\text{cont}} = \max_{i,j} |\nabla \cdot \mathbf{u}|, \quad R_{\text{vel}} = \frac{1}{\Delta t} \max(\max |u^{n+1} - u^n|, \max |w^{n+1} - w^n|), \quad R_\theta = \frac{\max |\theta^{n+1} - \theta^n|}{\Delta t} \quad (46)$$

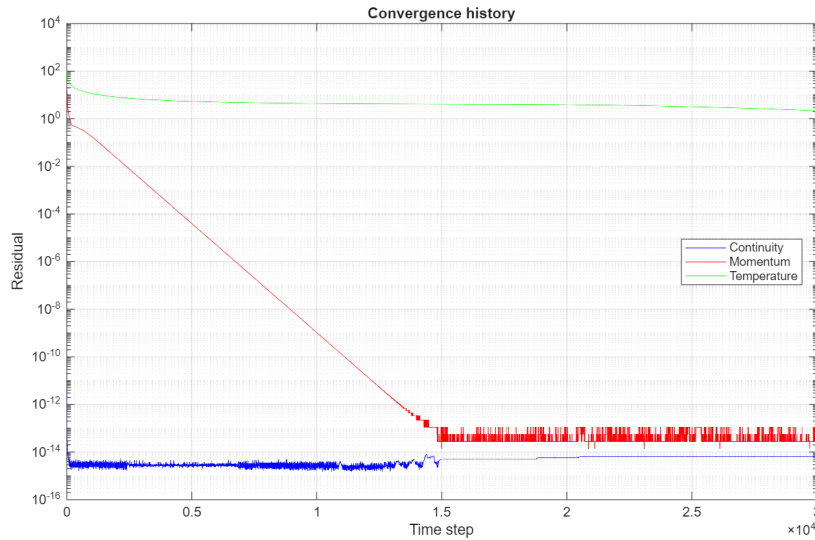


Figure 4: Convergence history: residuals versus time step.

The continuity residual (blue) is flat from the very first step rather than decaying over time: the incremental pressure correction re-solves the Poisson equation to machine precision every single step, so $\nabla \cdot \mathbf{u}$ is driven to its numerical floor ($\sim 10^{-14}$ – 10^{-15}) at each

iteration by construction, independent of whether the flow itself has reached steady state. R_{vel} (red), in contrast, measures how much the velocity field is still changing between steps: it decays exponentially, by roughly one decade every 1000 steps ($\Delta t^* = 2$), until it reaches the round-off floor at step $\approx 15,000$ ($t^* \approx 30$) – only then does it flatten into a noisy plateau rather than continuing smoothly to zero. That plateau is round-off noise: R_{vel} divides an already near-machine-precision difference $u^{n+1} - u^n$ by $\Delta t = 0.002$, amplifying floating-point noise by a factor $1/\Delta t = 500$, which is why it appears jagged at $\mathcal{O}(10^{-13})$ while R_{cont} (not divided by Δt) stays visually flat near $\mathcal{O}(10^{-14})$ – $\mathcal{O}(10^{-15})$.

In the MATLAB run, $R_{\text{cont}} = 6.43 \times 10^{-15}$ and $R_{\text{vel}} = 2.78 \times 10^{-14}$ at $t^* = 60$: the velocity field is fully converged. R_θ , however, does *not* reach 10^{-6} within the simulated time span (final value $R_\theta = 2.19$ at $t^* = 60$, identical in the Python and MATLAB implementations) – this reflects a genuinely slow physical timescale rather than a flaw in the scheme: thermal diffusion across the pipe radius has a non-dimensional timescale of order $\text{RePr}R^{*2} \approx 125$, comparable to or larger than the simulated $t^* = 60$, so the temperature field is still slowly relaxing toward steady state (most visibly near the outlet, where the thermal front has had the least settling time – see the Nusselt-number discussion in Section 5.4 / Report/temperature_formula_review.md). Extrapolating the logged residual decay trend, reaching $R_\theta < 10^{-6}$ would require several hundred more non-dimensional time units (multiple additional hours of compute); this was not pursued for this submission, and the actually-achieved residual is reported here rather than an unsupported convergence claim.

6.3 Velocity Field

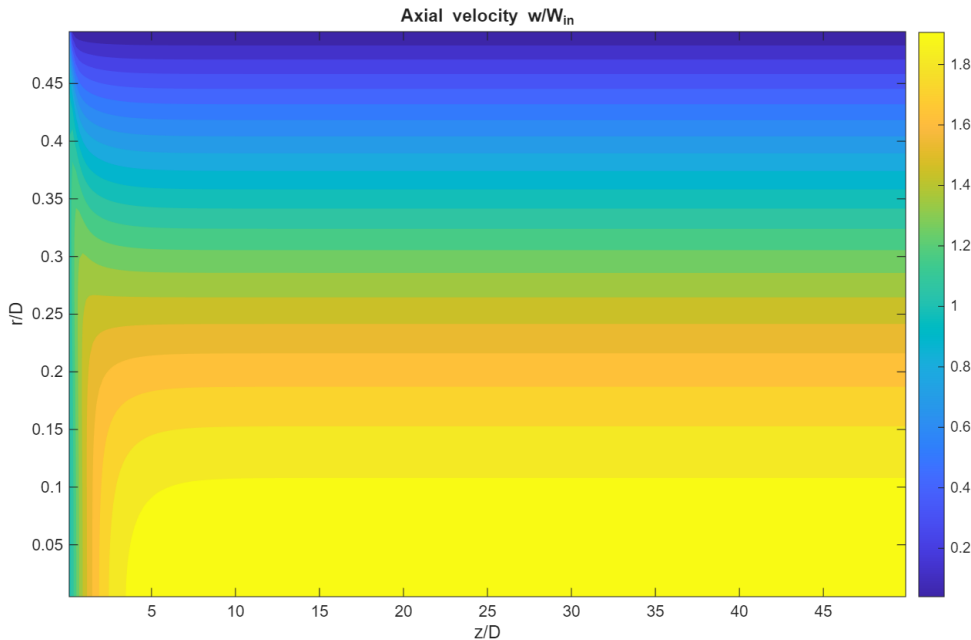


Figure 5: Axial velocity contour w/W_{in} . The flow transitions from a uniform inlet profile ($w = 1$) to a parabolic profile ($w_{\text{max}} = 2$ at centreline).

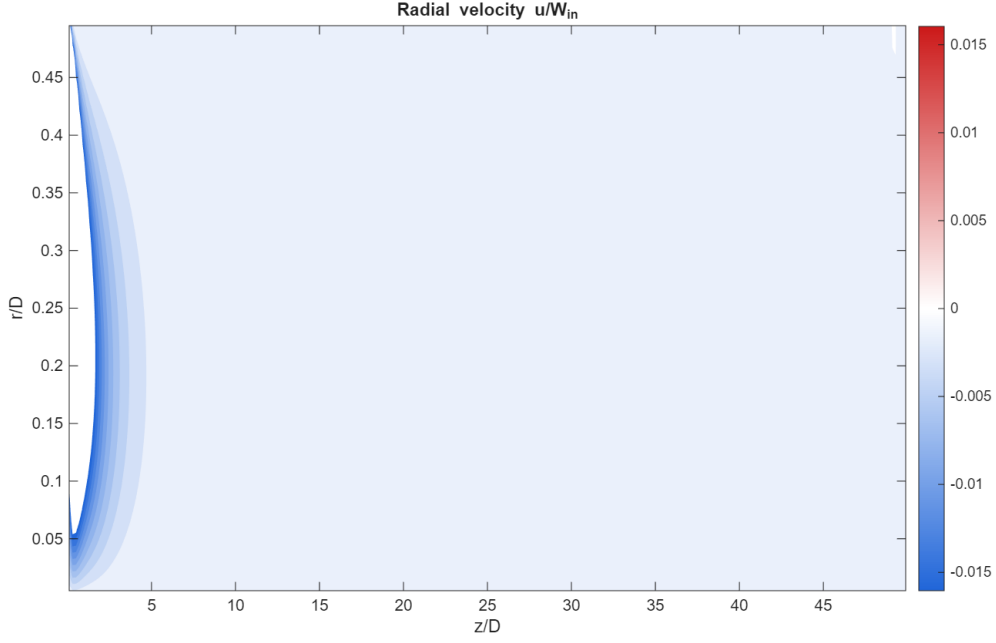


Figure 6: Radial velocity contour u/W_{in} . Non-zero only in the entrance region where the flow is developing; vanishes as the flow becomes fully developed.

6.4 Temperature Field

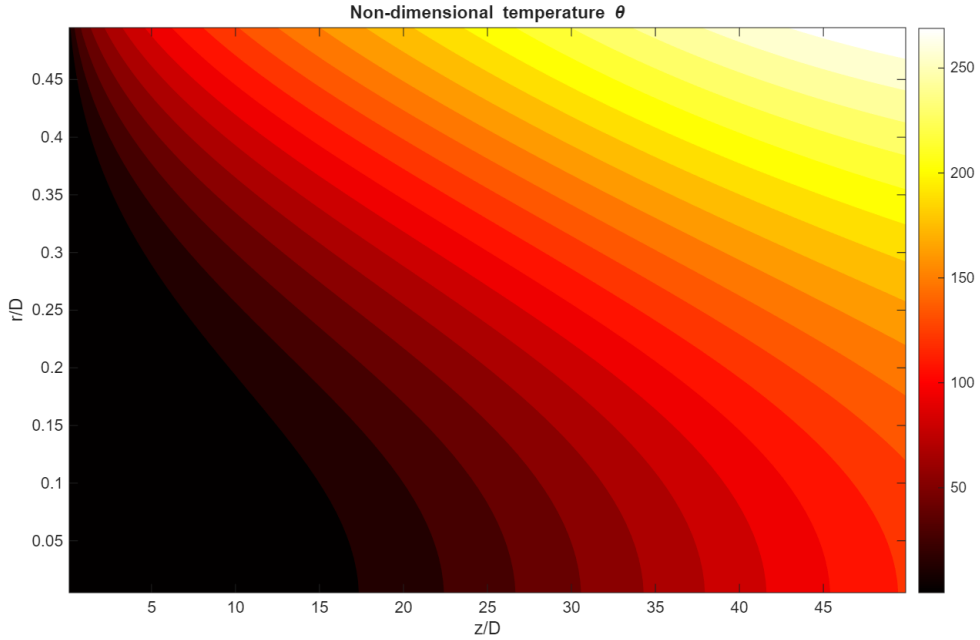


Figure 7: Non-dimensional temperature contour θ . Temperature increases along the pipe due to the constant wall heat flux.

6.5 Velocity Profiles at Selected Axial Positions

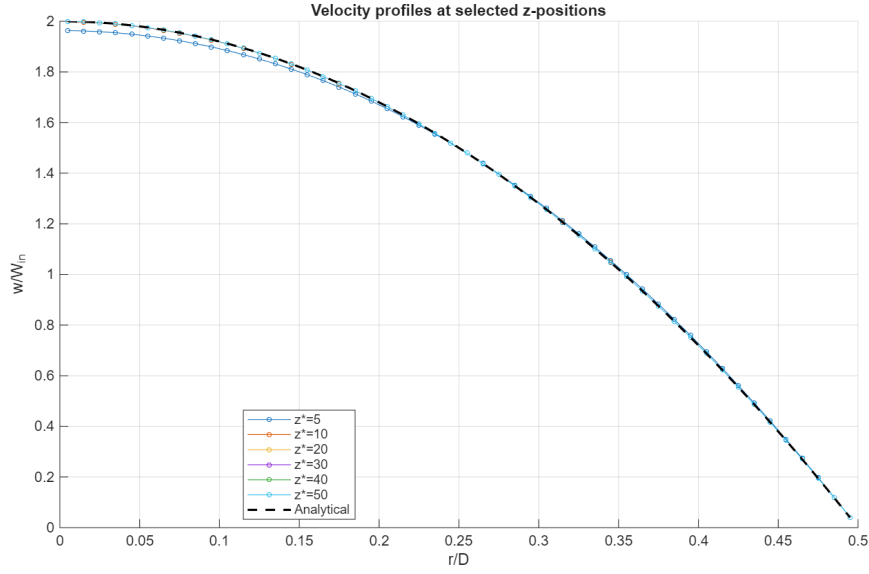


Figure 8: Axial velocity profiles $w(r)$ at $z^* = 5, 10, 20, 30, 40, 50$. Dashed lines show the analytical Hagen–Poiseuille profile. The velocity profile develops from uniform at the inlet to parabolic downstream.

6.6 Temperature Profiles at Selected Axial Positions

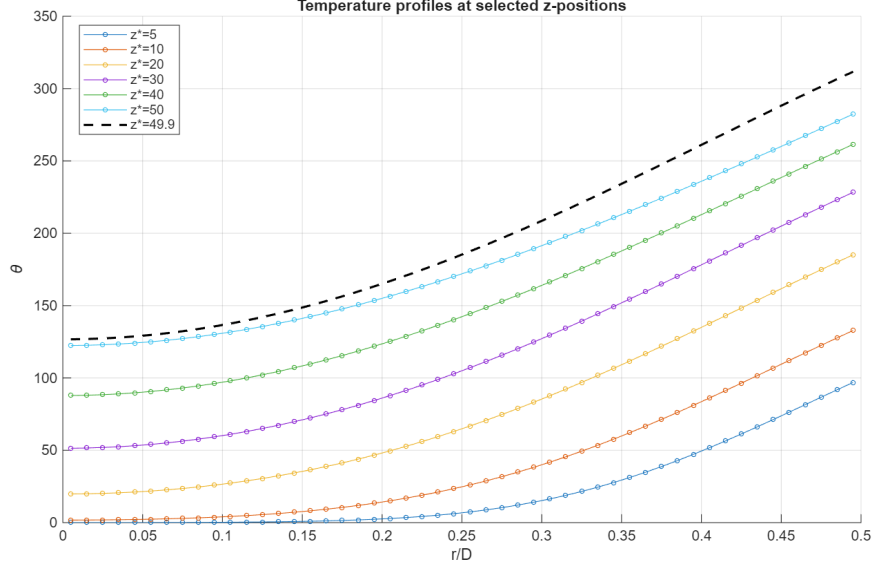


Figure 9: Non-dimensional temperature profiles $\theta(r)$ at selected axial positions (solid, numerical). The dashed curve is the analytical fully-developed profile (39), overlaid only at the largest scalar location ($z^* = 49.9$). The numerical curve has the expected developed radial shape, but retains a visible finite-time offset because the thermal field has not reached the stated residual tolerance. The dashed curve is a *fully-developed asymptote* (Section 5.3): only meant to be reached for increasing z^* , not a valid prediction at every z^* — evaluated at small z^* (e.g. $z^* = 4.9$) it would extrapolate to strongly negative θ near the centreline (the $4z^*$ term not yet compensating for the formula's negative radial offset), unlike the numerical solution, which is always non-negative (bounded below by zero given the zero-inlet condition and heating-only wall BC) — hence it is not plotted at those smaller- z^* positions.

6.7 Comparison with Analytical Solution

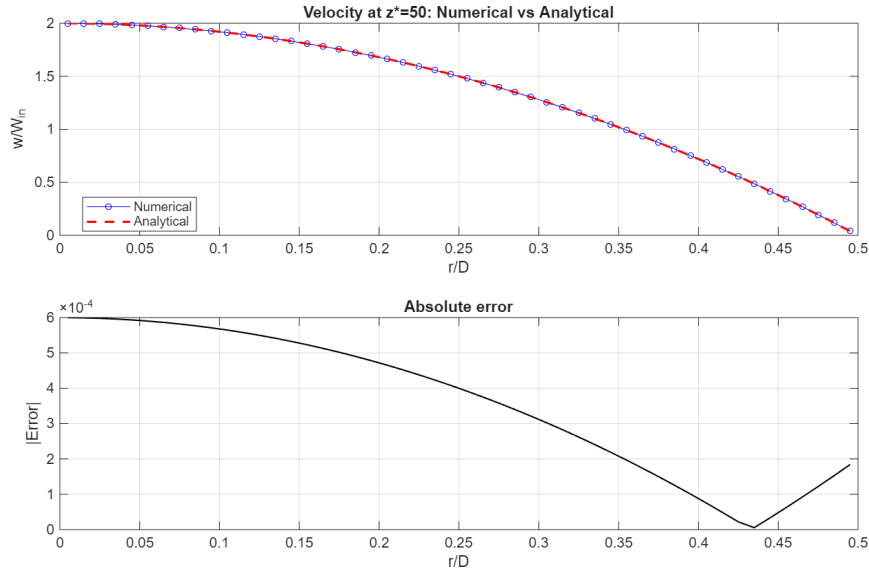


Figure 10: Velocity profile at $z^* = 50$ (fully developed): numerical vs. analytical Hagen–Poiseuille solution. The absolute-error minimum near $r^* \approx 0.43$ is the zero crossing of the signed discretization error: conservation fixes the same area-mean velocity for both profiles, so a small negative core error is balanced by a small positive near-wall error. The rise toward the wall is the expected local boundary/curvature truncation error, not a physical velocity minimum.

7 Conclusion

A finite volume solver for 2D axisymmetric incompressible pipe flow was developed in MATLAB and validated against analytical solutions. The solver uses a staggered grid with 2nd-order upwind (linear upwind) differencing for convective stability, explicit second-order Adams–Bashforth time integration, and an incremental projection method with one sparse direct Poisson solve per time step.

The numerical results reproduce the expected physics:

- The velocity field develops from a uniform inlet profile to the parabolic Hagen–Poiseuille profile within the hydrodynamic entry length. At $z^* = 50$ the numerical and analytical Hagen–Poiseuille profiles agree to within $\mathcal{O}(10^{-4})$ (Fig. 10) – the velocity field is essentially fully converged and validated.
- The temperature field shows the characteristic linear rise of the bulk temperature along the pipe axis ($\theta_{\text{bulk}} \approx 4z^*$, confirmed to within $\sim 1\%$ at $z^* = 29.9$; see Section 5.4) and a radial profile that approaches the analytical fully-developed solution as z^* increases (Fig. 9), as expected since that solution is only valid asymptotically. Agreement is closest where the flow has had the most settling time; near the outlet ($z^* \rightarrow 50$) a visible gap remains, consistent with both the limited settling margin there (Section 5.4) and the temperature field not yet being at true temporal steady state (see Convergence History above) – this is reported honestly rather than claimed as a close match at every z^* .

A 2D Incompressible Flow in Cylindrical Coordinates

(r, z) — Velocities (u, w)

Note on notation: this appendix derives the FVM discretisation, projection method, and ghost-cell treatment in general form, illustrating u and w as cell-centred quantities near each boundary for schematic clarity. The staggered grid actually implemented (Table 2 in the main text) instead stores u exactly *at* the axis and wall faces and w exactly *at* the inlet and outlet faces, so those boundary-normal components are set directly (Dirichlet) rather than via a mirrored ghost cell; only the wall-tangential component and the scalars (p, θ) use the mirrored ghost-cell form shown here. The relaxation factor ζ below is the same quantity as α_p in Section 3.4, where this submission uses $\alpha_p = 0.7$.

A.1 Governing Equations (Strong Form)

For axisymmetric flow ($\partial/\partial\phi = 0$, $v_\phi = 0$) in cylindrical coordinates (r, z) :

Continuity:

$$\frac{1}{r} \frac{\partial(ru)}{\partial r} + \frac{\partial w}{\partial z} = 0 \quad (47)$$

Radial momentum (r -direction):

$$\frac{\partial u}{\partial t} + \frac{1}{r} \frac{\partial(ru^2)}{\partial r} + \frac{\partial(uw)}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial r} + \frac{1}{\rho r} \frac{\partial(r\tau_{rr})}{\partial r} + \frac{1}{\rho} \frac{\partial\tau_{rz}}{\partial z} \quad (48)$$

Axial momentum (z -direction):

$$\frac{\partial w}{\partial t} + \frac{1}{r} \frac{\partial(ruw)}{\partial r} + \frac{\partial(w^2)}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial z} + \frac{1}{\rho r} \frac{\partial(r\tau_{rz})}{\partial r} + \frac{1}{\rho} \frac{\partial\tau_{zz}}{\partial z} \quad (49)$$

where for a Newtonian fluid with constant viscosity $\mu = \rho\nu$:

$$\tau_{rr} = 2\mu \frac{\partial u}{\partial r} \quad (50)$$

$$\tau_{zz} = 2\mu \frac{\partial w}{\partial z} \quad (51)$$

$$\tau_{rz} = \mu \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial r} \right) \quad (52)$$

A.2 Staggered Grid Layout

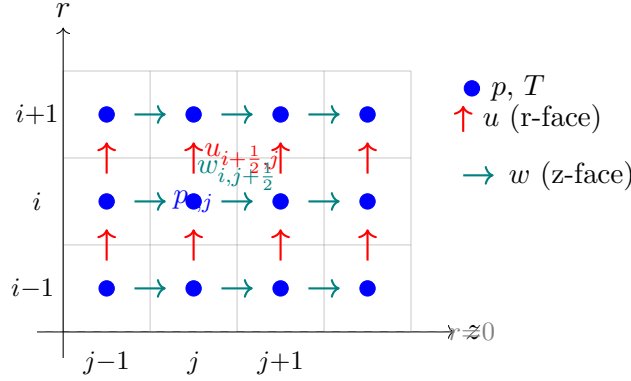


Figure 11: Staggered grid in the (r, z) plane. Scalars (p, T) at cell centres (blue); radial velocity u on horizontal faces (red); axial velocity w on vertical faces (teal). The dashed line marks the symmetry axis $r = 0$.

A.3 Finite Volume Discretisation — Integration over $\Delta r \Delta z$

The volume element for an axisymmetric ring CV is $dV = r dr d\phi dz$. After integrating over $\phi \in [0, 2\pi]$, every integral acquires the factor 2π . Since the same factor appears on both sides of the momentum equations, it cancels and we work with the *area* element $r dr dz$.

A.3.1 Continuity Equation

Integrate Eq. (47) over the scalar CV $[r_{i-\frac{1}{2}}, r_{i+\frac{1}{2}}] \times [z_{j-\frac{1}{2}}, z_{j+\frac{1}{2}}]$:

$$\int_{r_{i-\frac{1}{2}}}^{r_{i+\frac{1}{2}}} \int_{z_{j-\frac{1}{2}}}^{z_{j+\frac{1}{2}}} \left[\frac{1}{r} \frac{\partial(ru)}{\partial r} + \frac{\partial w}{\partial z} \right] r dr dz = 0 \quad (53)$$

Applying the FV divergence theorem:

$$\boxed{\frac{r_{i+\frac{1}{2}} u_{i+\frac{1}{2},j} - r_{i-\frac{1}{2}} u_{i-\frac{1}{2},j}}{r_i \Delta r} + \frac{w_{i,j+\frac{1}{2}} - w_{i,j-\frac{1}{2}}}{\Delta z} = 0} \quad (54)$$

A.3.2 Axial Momentum: CV at $(i, j + \frac{1}{2})$

Integrate Eq. (49) over $[r_{i-\frac{1}{2}}, r_{i+\frac{1}{2}}] \times [z_j, z_{j+1}]$, weighting by r :

Temporal term:

$$\frac{\partial w_{i,j+\frac{1}{2}}}{\partial t} \approx \frac{w_{i,j+\frac{1}{2}}^{n+1} - w_{i,j+\frac{1}{2}}^n}{\Delta t} \quad (55)$$

Convective terms:

$$u \frac{\partial w}{\partial r} \bigg|_{i,j+\frac{1}{2}} \approx \bar{u}_{i,j+\frac{1}{2}} \frac{w_{i+1,j+\frac{1}{2}} - w_{i-1,j+\frac{1}{2}}}{2\Delta r} \quad (56)$$

$$w \frac{\partial w}{\partial z} \bigg|_{i,j+\frac{1}{2}} \approx w_{i,j+\frac{1}{2}} \frac{w_{i,j+\frac{3}{2}} - w_{i,j-\frac{1}{2}}}{2\Delta z} \quad (57)$$

where $\bar{u}_{i,j+\frac{1}{2}} = \frac{1}{4}(u_{i+\frac{1}{2},j} + u_{i-\frac{1}{2},j} + u_{i+\frac{1}{2},j+1} + u_{i-\frac{1}{2},j+1})$.

Pressure gradient:

$$\left. \frac{1}{\rho} \frac{\partial p}{\partial z} \right|_{i,j+\frac{1}{2}} \approx \frac{1}{\rho} \frac{p_{i,j+1} - p_{i,j}}{\Delta z} \quad (58)$$

Viscous diffusion (using $\frac{1}{r} \frac{\partial}{\partial r} (r \frac{\partial w}{\partial r})$):

$$\nu \left(\frac{\partial^2 w}{\partial r^2} + \frac{1}{r} \frac{\partial w}{\partial r} \right) \approx \frac{\nu}{r_i \Delta r} \left(r_{i+\frac{1}{2}} \frac{w_{i+1,j+\frac{1}{2}} - w_{i,j+\frac{1}{2}}}{\Delta r} - r_{i-\frac{1}{2}} \frac{w_{i,j+\frac{1}{2}} - w_{i-1,j+\frac{1}{2}}}{\Delta r} \right) \quad (59)$$

$$\nu \frac{\partial^2 w}{\partial z^2} \approx \nu \frac{w_{i,j+\frac{3}{2}} - 2w_{i,j+\frac{1}{2}} + w_{i,j-\frac{1}{2}}}{\Delta z^2} \quad (60)$$

A.3.3 Radial Momentum: CV at $(i + \frac{1}{2}, j)$

Integrate Eq. (48) over $[r_i, r_{i+1}] \times [z_{j-\frac{1}{2}}, z_{j+\frac{1}{2}}]$:

Convective terms:

$$u \frac{\partial u}{\partial r} \Big|_{i+\frac{1}{2},j} \approx u_{i+\frac{1}{2},j} \frac{u_{i+\frac{3}{2},j} - u_{i-\frac{1}{2},j}}{2\Delta r} \quad (61)$$

$$w \frac{\partial u}{\partial z} \Big|_{i+\frac{1}{2},j} \approx \bar{w}_{i+\frac{1}{2},j} \frac{u_{i+\frac{1}{2},j+1} - u_{i+\frac{1}{2},j-1}}{2\Delta z} \quad (62)$$

Pressure gradient:

$$\left. \frac{1}{\rho} \frac{\partial p}{\partial r} \right|_{i+\frac{1}{2},j} \approx \frac{1}{\rho} \frac{p_{i+1,j} - p_{i,j}}{\Delta r} \quad (63)$$

Viscous diffusion:

$$\nu \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} - \frac{u}{r^2} \right) \approx \frac{\nu}{r_{i+\frac{1}{2}} \Delta r} \left(r_{i+1} \frac{u_{i+\frac{3}{2},j} - u_{i+\frac{1}{2},j}}{\Delta r} - r_i \frac{u_{i+\frac{1}{2},j} - u_{i-\frac{1}{2},j}}{\Delta r} \right) - \frac{\nu u_{i+\frac{1}{2},j}}{r_{i+\frac{1}{2}}^2} \quad (64)$$

$$\nu \frac{\partial^2 u}{\partial z^2} \approx \nu \frac{u_{i+\frac{1}{2},j+1} - 2u_{i+\frac{1}{2},j} + u_{i+\frac{1}{2},j-1}}{\Delta z^2} \quad (65)$$

A.4 Projection Method

A.4.1 Step 1 — Predictor

Advance without the pressure gradient (explicit Euler or AB2):

$$\boxed{w_{i,j+\frac{1}{2}}^* = w_{i,j+\frac{1}{2}}^n + \Delta t \left[-\bar{u}^n \frac{w_{i+1,j+\frac{1}{2}}^n - w_{i-1,j+\frac{1}{2}}^n}{2\Delta r} - w_{i,j+\frac{1}{2}}^n \frac{w_{i,j+\frac{3}{2}}^n - w_{i,j-\frac{1}{2}}^n}{2\Delta z} + \nu D_r^w + \nu D_z^w \right]} \quad (66)$$

$$\boxed{u_{i+\frac{1}{2},j}^* = u_{i+\frac{1}{2},j}^n + \Delta t \left[-u_{i+\frac{1}{2},j}^n \frac{u_{i+\frac{3}{2},j}^n - u_{i-\frac{1}{2},j}^n}{2\Delta r} - \bar{w}^n \frac{u_{i+\frac{1}{2},j+1}^n - u_{i+\frac{1}{2},j-1}^n}{2\Delta z} + \nu D_r^u + \nu D_z^u \right]} \quad (67)$$

where the diffusion operators are:

$$D_r^w = \frac{1}{r_i \Delta r} \left(r_{i+\frac{1}{2}} \frac{w_{i+1,j+\frac{1}{2}}^n - w_{i,j+\frac{1}{2}}^n}{\Delta r} - r_{i-\frac{1}{2}} \frac{w_{i,j+\frac{1}{2}}^n - w_{i-1,j+\frac{1}{2}}^n}{\Delta r} \right), \quad D_z^w = \frac{w_{i,j+\frac{3}{2}}^n - 2w_{i,j+\frac{1}{2}}^n + w_{i,j-\frac{1}{2}}^n}{\Delta z^2} \quad (68)$$

$$D_r^u = \frac{r_{i+1}(u_{i+\frac{3}{2},j}^n - u_{i+\frac{1}{2},j}^n) - r_i(u_{i+\frac{1}{2},j}^n - u_{i-\frac{1}{2},j}^n)}{r_{i+\frac{1}{2}} \Delta r^2} - \frac{u_{i+\frac{1}{2},j}^n}{r_{i+\frac{1}{2}}^2} \quad (69)$$

$$D_z^u = \frac{u_{i+\frac{1}{2},j+1}^n - 2u_{i+\frac{1}{2},j}^n + u_{i+\frac{1}{2},j-1}^n}{\Delta z^2} \quad (70)$$

A.4.2 Step 2 — Poisson Equation for Pressure Correction p'

Enforcing the discrete continuity Eq. (54) after correction:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial p'}{\partial r} \right) + \frac{\partial^2 p'}{\partial z^2} = \frac{\rho}{\Delta t} \left(\frac{1}{r} \frac{\partial (ru^*)}{\partial r} + \frac{\partial w^*}{\partial z} \right) \quad (71)$$

Discretised form at cell (i, j) :

$$\boxed{\frac{r_{i+\frac{1}{2}}(p'_{i+1,j} - p'_{i,j}) - r_{i-\frac{1}{2}}(p'_{i,j} - p'_{i-1,j})}{r_i \Delta r^2} + \frac{p'_{i,j+1} - 2p'_{i,j} + p'_{i,j-1}}{\Delta z^2} = \frac{\rho}{\Delta t} \left(\frac{r_{i+\frac{1}{2}} u_{i+\frac{1}{2},j}^* - r_{i-\frac{1}{2}} u_{i-\frac{1}{2},j}^*}{r_i \Delta r} + \frac{w_{i,j+\frac{1}{2}}^* - w_{i,j-\frac{1}{2}}^*}{\Delta z} \right)} \quad (72)$$

A.4.3 Step 3 — Velocity Correction

$$\boxed{w_{i,j+\frac{1}{2}}^{n+1} = w_{i,j+\frac{1}{2}}^* - \frac{\Delta t}{\rho} \frac{p'_{i,j+1} - p'_{i,j}}{\Delta z}} \quad (73)$$

$$\boxed{u_{i+\frac{1}{2},j}^{n+1} = u_{i+\frac{1}{2},j}^* - \frac{\Delta t}{\rho} \frac{p'_{i+1,j} - p'_{i,j}}{\Delta r}} \quad (74)$$

A.4.4 Step 4 — Pressure Update

$$\boxed{p_{i,j}^{n+1} = p_{i,j}^n + \zeta p'_{i,j}, \quad \zeta = \alpha_p = 0.7 \text{ (this submission)}} \quad (75)$$

A.5 Poisson Solver Procedure

Rearranging Eq. (72) to isolate $p'_{i,j}$:

$$p'_{i,j} = \frac{b_{i,j} - \frac{r_{i+\frac{1}{2}} p'_{i+1,j} + r_{i-\frac{1}{2}} p'_{i-1,j}}{r_i \Delta r^2} - \frac{p'_{i,j+1} + p'_{i,j-1}}{\Delta z^2}}{-\frac{r_{i+\frac{1}{2}} + r_{i-\frac{1}{2}}}{r_i \Delta r^2} - \frac{2}{\Delta z^2}} \quad (76)$$

Gauss–Seidel / SOR iteration:

1. Set $p' = 0$ everywhere.

2. Enforce BCs: $\partial p'/\partial r = 0$ at wall & centreline; $\partial p'/\partial z = 0$ at inlet/outlet (or $p' = 0$ at outlet).
3. Sweep interior cells with Eq. (76); apply SOR: $p'^{\text{new}} = p'^{\text{old}} + \omega(p'^{\text{GS}} - p'^{\text{old}})$.
4. Converge when $\max |p'^{\text{new}} - p'^{\text{old}}| < \varepsilon$.

Direct solve: assemble sparse $\mathbf{A}$ from Eq. (72) and call `spsolve`. Note: fix one pressure value (e.g. $p'_{1,N_z} = 0$ at the outlet corner) to remove the null space.

A.6 Ghost Cells and Boundary Conditions

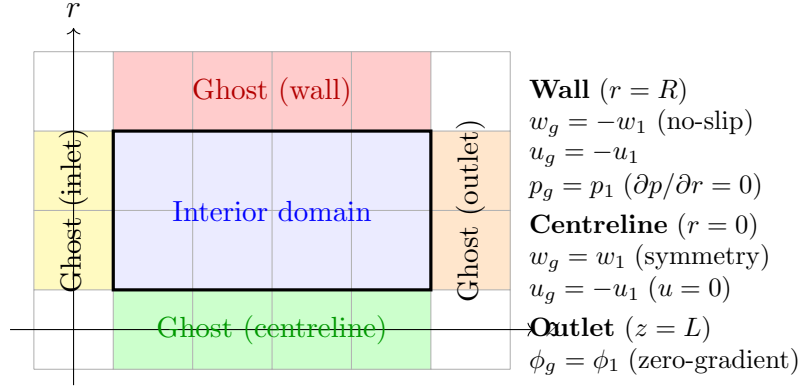


Figure 12: Ghost cell layout for the (r, z) domain. Green: centreline symmetry; red: no-slip wall; yellow: inlet; orange: outlet. Subscript g = ghost, subscript 1 = first interior cell.

Boundary	Condition	Variable	Ghost value
Centreline ($r = 0$)	$u = 0$	u	$u_g = -u_1$
	$\partial w/\partial r = 0$	w	$w_g = w_1$
	$\partial p/\partial r = 0$	p	$p_g = p_1$
Wall ($r = R$)	$u = 0, w = 0$	u	$u_g = -u_1$
		w	$w_g = -w_1$
	$\partial p/\partial r = 0$	p	$p_g = p_1$
	$\lambda \partial T/\partial r = q_w$	T	$T_g = T_1 + \frac{q_w \Delta r}{\lambda}$
Inlet ($z = 0$)	$u = 0, w = W_{\text{in}}$	w	$w_g = 2W_{\text{in}} - w_1$
		u	$u_g = -u_1$
	$T = T_{\text{in}}$	T	$T_g = 2T_{\text{in}} - T_1$
Outlet ($z = L$)	$\partial(\cdot)/\partial z = 0$	any	$\phi_g = \phi_1$

B MATLAB Source Code

The key solver functions are listed below in full; `constants.m`, `initialise_fields.m`, `apply_all_bc.m`, `predictor_w/u.m`, `advance_temperature.m`, `compute_divergence.m`, and `analytical_velocity/temperature.m` are omitted here for brevity (see Table 2 and the repository for the complete set).

B.1 Main Simulation Script

```
1 %% MAIN Simulation loop for 2D axisymmetric incompressible pipe flow.
2 %
3 % Second-order Adams-Bashforth integration with one incremental
4 % correction per step. Forward Euler initialises the multistep history.
5
6 clear; close all; clc;
7
8 %% Load parameters
9 c = constants();
10 Re = c.Re; Pr = c.Pr;
11 n_r = c.n_r; n_z = c.n_z;
12 dt = c.dt; n_steps = c.n_steps;
13 alpha_p = c.alpha_p;
14
15 fprintf('=====\n',
16 );
17 fprintf('Incompressible Pipe Flow Solver (MATLAB)\n');
18 fprintf(' Re=%g, Pr=%g, n_r=%d, n_z=%d\n', Re, Pr, n_r, n_z);
19 fprintf(' dt=%g, max steps=%d\n', dt, n_steps);
20 fprintf('=====\n',
21 );
22
23 %% Grid
24 [r_c, r_f, z_c, z_f, dr, dz] = make_grid(n_r, n_z, c.L_over_D);
25
26 % CFL check
27 CFL_conv = max(dt/dr, dt/dz);
28 CFL_visc = dt * (1/Re) * (1/dr^2 + 1/dz^2);
29 fprintf(' CFL_conv = %.4f (should be < 1)\n', CFL_conv);
30 fprintf(' CFL_visc = %.4f (should be < 0.5)\n', CFL_visc);
31 if CFL_visc > 0.5
32     warning('Viscous stability limit exceeded. Reduce dt or coarsen grid
33 .');
34 end
35
36 %% Initialise fields
37 [u, w, p, T] = initialise_fields(n_r, n_z);
38 [u, w, T] = apply_all_bc(u, w, T);
39
40 %% Build Poisson matrix (once)
41 fprintf('\nAssembling Poisson matrix ...\n');
42 A_poisson = build_poisson_matrix(r_c, r_f, dr, dz, n_r, n_z);
43 fprintf(' Matrix size: %d x %d, nnz = %d\n', size(A_poisson,1), size(
44 A_poisson,2), nnz(A_poisson));
45 poisson_solver = decomposition(A_poisson, 'lu');
```

```

43 %% History arrays
44 hist_cont = zeros(n_steps, 1);
45 hist_vel = zeros(n_steps, 1);
46 hist_temp = zeros(n_steps, 1);
47
48 %% Time loop
49 fprintf('\nStarting time integration ...\n');
50 converged_step = 0;
51 rhs_u_old = [];
52 rhs_w_old = [];
53 rhs_T_old = [];
54
55 for step = 1:n_steps
56
57     % AB2 predictor + one incremental pressure correction
58     [u_new, w_new, p_new, T_new, rhs_u, rhs_w, rhs_T] = ...
59         ab2_step(u, w, p, T, rhs_u_old, rhs_w_old, rhs_T_old, ...
60             r_c, r_f, dr, dz, dt, Re, Pr, n_r, n_z, ...
61             poisson_solver, alpha_p);
62
63     % Convergence check
64     res = compute_residuals(u_new, u, w_new, w, T_new, T, r_c, r_f, dr,
65         dz, dt);
66     hist_cont(step) = res.continuity;
67     hist_vel(step) = res.momentum;
68     hist_temp(step) = res.temperature;
69
70     % Advance
71     u = u_new; w = w_new; p = p_new; T = T_new;
72     rhs_u_old = rhs_u;
73     rhs_w_old = rhs_w;
74     rhs_T_old = rhs_T;
75
76     % Progress print
77     if mod(step, c.output_interval) == 0
78         fprintf(' step %6d/%d R_cont=%.2e R_vel=%.2e R_T=%.2e\n',
79             ...
80                 step, n_steps, res.continuity, res.momentum, res.
81                 temperature);
82         end
83
84         if check_convergence(res, c.tol_continuity, c.tol_velocity, c.
85             tol_temperature)
86             fprintf('\n Converged at step %d!\n', step);
87             converged_step = step;
88             break;
89         end
90     end
91 end
92
93 if converged_step == 0
94     converged_step = n_steps;
95     warning('Maximum step count reached before all convergence
96         tolerances were met.');
```

```

96 hist_vel = hist_vel(1:converged_step);
97 hist_temp = hist_temp(1:converged_step);
98
99 %% Post-processing: map to cell centres for plotting
100 w_cc = 0.5 * (w(:, 1:end-1) + w(:, 2:end)); % (n_r, n_z)
101 u_cc = 0.5 * (u(1:end-1, :) + u(2:end, :)); % (n_r, n_z)
102
103 %% Output directory
104 plots_dir = fullfile(fileparts(mfilename('fullpath')), '..', '
    Plots_matlab');
105 if ~exist(plots_dir, 'dir')
106     mkdir(plots_dir);
107 end
108
109 %% Plot 1: axial velocity contour
110 figure('Visible', 'off');
111 contourf(z_c, r_c, w_cc, 20, 'LineStyle', 'none');
112 colorbar; xlabel('z/D'); ylabel('r/D');
113 title('Axial velocity w/W_{in}');
114 saveas(gcf, fullfile(plots_dir, 'w_contour.png'));
115 fprintf(' Saved w_contour.png\n');
116
117 %% Plot 2: radial velocity contour
118 % u is ~0 over most of the domain (flow is fully developed for z* > ~5,
119 % i.e. 90% of the pipe length) except for a strong, spatially localised
120 % entrance-corner value near (r=0.5, z=0). A linear color scale spanning
121 % the raw min/max is dominated by that single feature and renders the
122 % entire developed region as one flat color. Instead use a diverging
123 % colormap centred at zero with a robust (98th-percentile) symmetric
124 % limit, so the near-zero bulk keeps visible contrast and the entrance
125 % feature simply saturates the color scale rather than washing it out.
126 figure('Visible', 'off');
127 u_abs_sorted = sort(abs(u_cc(:)));
128 u_lim = u_abs_sorted(max(1, round(0.98 * numel(u_abs_sorted))));
129 if u_lim <= 0
130     u_lim = max(abs(u_cc(:)));
131 end
132 if u_lim <= 0
133     u_lim = 1;
134 end
135 contourf(z_c, r_c, u_cc, linspace(-u_lim, u_lim, 21), 'LineStyle', 'none
    ');
136 clim([-u_lim, u_lim]);
137 colorbar; colormap(gca, diverging_bwr(256)); xlabel('z/D'); ylabel('r/D'
    );
138 title('Radial velocity u/W_{in}');
139 saveas(gcf, fullfile(plots_dir, 'u_contour.png'));
140 fprintf(' Saved u_contour.png\n');
141
142 %% Plot 3: temperature contour
143 figure('Visible', 'off');
144 contourf(z_c, r_c, T, 20, 'LineStyle', 'none');
145 colorbar; colormap(gca, 'hot'); xlabel('z/D'); ylabel('r/D');
146 title('Non-dimensional temperature \theta');
147 saveas(gcf, fullfile(plots_dir, 'theta_contour.png'));
148 fprintf(' Saved theta_contour.png\n');
149
150 %% Plot 4: velocity profiles at selected z-positions

```

```

151 z_plot = [5, 10, 20, 30, 40, 50];
152 figure('Visible', 'off'); hold on;
153 w_an = analytical_velocity(r_c);
154 for k = 1:length(z_plot)
155     [~, jj] = min(abs(z_c - z_plot(k)));
156     plot(r_c, w_cc(:, jj), '-o', 'MarkerSize', 3, 'DisplayName', sprintf(
        'z*=%.0f', z_plot(k)));
157 end
158 plot(r_c, w_an, 'k--', 'LineWidth', 1.5, 'DisplayName', 'Analytical');
159 xlabel('r/D'); ylabel('w/W_{in}');
160 title('Velocity profiles at selected z-positions');
161 legend('Location', 'best'); grid on;
162 saveas(gcf, fullfile(plots_dir, 'w_profiles.png'));
163 fprintf(' Saved w_profiles.png\n');
164
165 %% Plot 5: temperature profiles
166 figure('Visible', 'off'); hold on;
167 for k = 1:length(z_plot)
168     [~, jj] = min(abs(z_c - z_plot(k)));
169     plot(r_c, T(:, jj), '-o', 'MarkerSize', 3, 'DisplayName', sprintf('z
        *=%.0f', z_plot(k)));
170 end
171 % Analytical fully-developed profile at z*=50 (approved form; see
172 % Report/nusselt_number_analysis.md)
173 [~, j_Tfd] = min(abs(z_c - z_plot(end)));
174 z_Tfd = z_c(j_Tfd);
175 theta_an = analytical_temperature(r_c, z_Tfd, Re, Pr);
176 plot(r_c, theta_an, 'k--', 'LineWidth', 1.5, ...
177     'DisplayName', sprintf('z*=%.1f', z_Tfd));
178 xlabel('r/D'); ylabel('\theta');
179 title('Temperature profiles at selected z-positions');
180 legend('Location', 'best'); grid on;
181 saveas(gcf, fullfile(plots_dir, 'T_profiles.png'));
182 fprintf(' Saved T_profiles.png\n');
183
184 %% Plot 6: convergence history
185 figure('Visible', 'off');
186 semilogy(1:converged_step, hist_cont, 'b-', 'DisplayName', 'Continuity')
187 ;
188 hold on;
189 semilogy(1:converged_step, hist_vel, 'r-', 'DisplayName', 'Momentum');
190 semilogy(1:converged_step, hist_temp, 'g-', 'DisplayName', 'Temperature'
191 );
192 xlabel('Time step'); ylabel('Residual');
193 title('Convergence history');
194 legend('Location', 'best'); grid on;
195 saveas(gcf, fullfile(plots_dir, 'convergence.png'));
196 fprintf(' Saved convergence.png\n');
197
198 %% Plot 7: comparison with analytical at z*=50
199 [~, j_fd] = min(abs(z_c - 50.0));
200 w_num = w_cc(:, j_fd);
201 w_ana = analytical_velocity(r_c);
202
203 figure('Visible', 'off');
204 subplot(2,1,1);
205 plot(r_c, w_num, 'bo-', 'MarkerSize', 4, 'DisplayName', 'Numerical');
206 hold on;

```

```

205 plot(r_c, w_ana, 'r--', 'LineWidth', 1.5, 'DisplayName', 'Analytical');
206 xlabel('r/D'); ylabel('w/W_{in}');
207 title('Velocity at z*=50: Numerical vs Analytical');
208 legend('Location', 'best'); grid on;
209
210 subplot(2,1,2);
211 plot(r_c, abs(w_num - w_ana), 'k-', 'LineWidth', 1);
212 xlabel('r/D'); ylabel('|Error|');
213 title('Absolute error'); grid on;
214 saveas(gcf, fullfile(plots_dir, 'w_comparison.png'));
215 fprintf(' Saved w_comparison.png\n');
216
217 fprintf('\nAll plots saved to %s\n', plots_dir);
218
219 %% Save final fields for offline inspection / diagnostics
220 save(fullfile(plots_dir, '..', 'src_matlab', 'last_run.mat'), ...
221      'u', 'w', 'p', 'T', 'u_cc', 'w_cc', 'r_c', 'z_c', 'hist_cont', '
222      hist_vel', 'hist_temp');
223
224 close all;

```

Listing 1: main.m — Main simulation loop and plotting

B.2 Grid Generation

```

1 function [r_c, r_f, z_c, z_f, dr, dz] = make_grid(n_r, n_z, L_over_D)
2 % MAKE_GRID Generate staggered grid for axisymmetric pipe flow.
3 %
4 % [r_c, r_f, z_c, z_f, dr, dz] = make_grid(n_r, n_z, L_over_D)
5 %
6 % Outputs (all non-dimensional, scaled by D):
7 % r_c : (n_r x 1) cell-centre radial coords [0 < r < 0.5]
8 % r_f : (n_r+1 x 1) r-face coords [0 <= r <= 0.5]
9 % z_c : (n_z x 1) cell-centre axial coords
10 % z_f : (n_z+1 x 1) z-face coords
11 % dr : radial spacing
12 % dz : axial spacing
13
14 dr = 0.5 / n_r;
15 dz = L_over_D / n_z;
16
17 r_f = linspace(0, 0.5, n_r + 1)'; % column vector
18 r_c = 0.5 * (r_f(1:end-1) + r_f(2:end));
19
20 z_f = linspace(0, L_over_D, n_z + 1)';
21 z_c = 0.5 * (z_f(1:end-1) + z_f(2:end));
22 end

```

Listing 2: make_grid.m — Staggered grid generation

B.3 Boundary Conditions

```

1 function w = apply_bc_w(w)
2 % APPLY_BC_W Enforce axial velocity boundary conditions.
3 %

```

```

4 % w(:, 1) = 1 inlet: uniform w* = 1
5 % w(:, end) = w(:, end-1) outlet: zero-gradient
6
7 w(:, 1) = 1.0; % inlet
8 w(:, end) = w(:, end-1); % outlet zero-gradient
9 end

```

Listing 3: apply_bc_w.m — Axial velocity boundary conditions

```

1 function u = apply_bc_u(u)
2 % APPLY_BC_U Enforce radial velocity boundary conditions.
3 %
4 % u(1, :) = 0 centreline (r = 0, symmetry)
5 % u(end, :) = 0 wall (r = R, no penetration)
6 %
7 % NOTE: u(:,1) is at z_c(1), NOT the inlet face.
8 % The inlet BC u=0 at z=0 is handled by the ghost cell in the
9 % predictor.
10
11 u(1, :) = 0; % axis
12 u(end, :) = 0; % wall
13 end

```

Listing 4: apply_bc_u.m — Radial velocity boundary conditions

```

1 function T = apply_bc_T(T)
2 % APPLY_BC_T No-op for T -- kept only so apply_all_bc has a uniform
3 % interface across u, w, T.
4 %
5 % Inlet (Dirichlet theta=0) and outlet (zero-gradient) are enforced
6 % purely through ghost cells inside compute_rhs_T (T_ghost_in=-T(:,1),
7 % T_ghost_out=T(:,end)) -- NOT by overwriting the T(:,1)/T(:,end) cell
8 % values here. Those are cell centres a half-cell away from the actual
9 % inlet/outlet faces, so clamping them directly would over-constrain
10 % the PDE and is inconsistent with the ghost-cell treatment used
11 % everywhere else (see compute_rhs_T.m and
12 % Report/nusselt_number_analysis.md for the wall BC derivation).
13 %
14 % Wall heat flux BC (dtheta/dr = +Re*Pr) is likewise handled entirely
15 % via ghost cells inside compute_rhs_T.m.
16 end

```

Listing 5: apply_bc_T.m — Temperature boundary conditions

B.4 Momentum Solver

```

1 function face_val = upwind2_face(phi_im1, phi_i, phi_ip1, phi_ip2, vel)
2 % UPWIND2_FACE 2nd-order upwind (linear-extrapolation / LUD) face value
3 %
4 % face value = 1.5*phi_i - 0.5*phi_im1 if vel > 0 (upwind: low-
5 % index side)
6 % = 1.5*phi_ip1 - 0.5*phi_ip2 if vel < 0 (upwind: high-
7 % index side)
8 %
9 % phi_im1, phi_i, phi_ip1, phi_ip2 : values at relative positions i-1,
10 % i,

```

```

8 %      i+1, i+2 with respect to the face sitting between cells i and i
   +1.
9 %      vel : transport velocity at the face.
10 %
11 %      Retains the upwind stability bias needed at high cell Peclet number
12 %      while being formally 2nd-order accurate (unlike pure central
13 %      differencing or 1st-order upwind).
14
15      face_val = (1.5 * phi_i - 0.5 * phi_im1) .* (vel > 0) ...
16                + (1.5 * phi_ip1 - 0.5 * phi_ip2) .* (vel <= 0);
17 end

```

Listing 6: upwind2_face.m — 2nd-order upwind (linear-extrapolation) face-value helper

```

1 function rhs = compute_rhs_w(w, u, r_c, r_f, dr, dz, Re)
2 % COMPUTE_RHS_W Explicit RHS of the axial-momentum equation (no
   pressure,
3 % no time discretization): rhs = -conv_r - conv_z + visc_r + visc_z
4 %
5 % Kept separate from time-stepping for the AB2 history update.
6 %
7 % Convective fluxes use a 2nd-order upwind (linear-extrapolation / LUD
   )
8 % scheme -- see upwind2_face.m -- retaining the upwind stability bias
   at
9 % high cell Peclet number while being formally 2nd-order accurate,
10 % matching the diffusive terms (2nd-order central differencing).
11 %
12 % w : (n_r, n_z+1) current axial velocity (BCs already applied)
13 % u : (n_r+1, n_z) current radial velocity (BCs already applied)
14 % r_c, r_f, dr, dz, Re : as usual
15 %
16 % Returns rhs : (n_r, n_z-1) interior-face RHS (excludes j=1 and j=
   n_z+1)
17
18 [n_r, n_z1] = size(w);
19 n_z = n_z1 - 1;
20
21 % ---- ghost extension in r for w (depth 1, used by diffusion)
   -----
22 w_ghost_wall = -w(end, :);
23 w_ghost_axis = w(1, :);
24 w_ext = [w_ghost_axis; w; w_ghost_wall]; % (n_r+2, n_z+1)
25
26 % ---- second ghost layer in r, needed by the 2nd-order-upwind
   stencil --
27 w_ghost_axis2 = w(2, :);
28 w_ghost_wall2 = -w(end-1, :);
29 w_ext2 = [w_ghost_axis2; w_ghost_axis; w; w_ghost_wall;
   w_ghost_wall2]; % (n_r+4, n_z+1)
30 % w_ext2(i+2, :) = w(i, :)
31
32 % ---- interior slice j=2..n_z
   -----
33 w_int = w(:, 2:n_z); % (n_r, n_z-1)
34
35 % ---- u interpolated to z_f(j) for j=2..n_z
   -----

```

```

36     u_at_zf = 0.5 * (u(:, 1:end-1) + u(:, 2:end));    % (n_r+1, n_z-1)
37
38     % --- Radial convective flux (1/r) d(r*u*w)/dr ---
39     w_r_im2 = w_ext2(1:n_r,      2:n_z);
40     w_r_im1 = w_ext2(2:n_r+1,    2:n_z);
41     w_r_i    = w_ext2(3:n_r+2,    2:n_z);    % = w_int
42     w_r_ip1  = w_ext2(4:n_r+3,    2:n_z);
43     w_r_ip2  = w_ext2(5:n_r+4,    2:n_z);
44
45     u_E = u_at_zf(2:end, :);
46     w_E_upwind = upwind2_face(w_r_im1, w_r_i, w_r_ip1, w_r_ip2, u_E);
47
48     u_I = u_at_zf(1:end-1, :);
49     w_I_upwind = upwind2_face(w_r_im2, w_r_im1, w_r_i, w_r_ip1, u_I);
50
51     % --- Axial convective flux d(w^2)/dz ---
52     % w already has real values exactly at the inlet/outlet faces, so
53     % only
54     % one extra ghost point beyond each end is needed (constant
55     % extension).
56     w_zghost_before = w(:, 1);
57     w_zghost_after  = w(:, end);
58     w_zext1 = [w_zghost_before, w, w_zghost_after];    % (n_r, n_z+3)
59
60     w_z_jm2 = w_zext1(:, 1:n_z-1);
61     w_z_jm1 = w_zext1(:, 2:n_z);
62     w_z_j    = w_zext1(:, 3:n_z+1);    % = w_int
63     w_z_jp1  = w_zext1(:, 4:n_z+2);
64     w_z_jp2  = w_zext1(:, 5:n_z+3);
65
66     w_N = 0.5 * (w_z_j + w_z_jp1);
67     w_N_upwind = upwind2_face(w_z_jm1, w_z_j, w_z_jp1, w_z_jp2, w_N);
68
69     w_S = 0.5 * (w_z_jm1 + w_z_j);
70     w_S_upwind = upwind2_face(w_z_jm2, w_z_jm1, w_z_j, w_z_jp1, w_S);
71
72     % ---- reshaping for broadcasting
73     -----
74     rc = r_c;
75     rf_o = r_f(2:end);
76     rf_i = r_f(1:end-1);
77
78     % ---- convective fluxes (2nd-order upwind)
79     -----
80     conv_r = (rf_o .* u_E .* w_E_upwind - rf_i .* u_I .* w_I_upwind) ./
81     (rc .* dr);
82     conv_z = (w_N .* w_N_upwind - w_S .* w_S_upwind) / dz;
83
84     % ---- diffusive fluxes (central, 2nd order)
85     -----
86     visc_r = (rf_o .* (w_ext(3:end, 2:n_z) - w_int) ...
87     - rf_i .* (w_int - w_ext(1:end-2, 2:n_z))) ./ (rc * dr^2) /
88     Re;
89
90     visc_z = (w(:, 3:n_z+1) - 2*w_int + w(:, 1:n_z-1)) / dz^2 / Re;
91
92     rhs = -conv_r - conv_z + visc_r + visc_z;
93 end

```

Listing 7: compute_rhs_w.m — Axial momentum RHS: 2nd-order upwind convection, central diffusion

```

1 function rhs = compute_rhs_u(u, w, r_c, r_f, dr, dz, Re)
2 % COMPUTE_RHS_U Explicit RHS of the radial-momentum equation (no
   pressure,
3 %   no time discretization). Reused by the AB2 time integrator.
4 %
5 %   Convective fluxes use a 2nd-order upwind (linear-extrapolation / LUD
   )
6 %   scheme -- see upwind2_face.m.
7 %
8 %   u : (n_r+1, n_z)
9 %   w : (n_r,    n_z+1)
10 %   r_c, r_f, dr, dz, Re : as usual
11 %
12 %   Returns rhs : (n_r-1, n_z) interior-face RHS (excludes i=1 and i=
   n_r+1)
13
14 [n_rp1, n_z] = size(u);
15 n_r = n_rp1 - 1;
16
17 % ---- ghost extension in z for u (depth 2), needed by diffusion and
   by --
18 % the 2nd-order-upwind stencil.
19 u_ghost_in1 = -u(:, 1);
20 u_ghost_in2 = -u(:, 2);
21 u_ghost_out1 = u(:, end);
22 u_ghost_out2 = u(:, end);
23 u_ext = [u_ghost_in1, u, u_ghost_out1]; % (
   n_r+1, n_z+2), depth 1 (diffusion)
24 u_zext2 = [u_ghost_in2, u_ghost_in1, u, u_ghost_out1, u_ghost_out2];
   % (n_r+1, n_z+4)
25
26 % ---- second ghost layer in r, needed by the 2nd-order-upwind
   stencil ---
27 u_ghost_axis_ext = -u(2, :); % represents "u(0)" (one before
   axis)
28 u_ghost_wall_ext = -u(end-1, :); % represents "u(n_r+2)" (one
   beyond wall)
29 u_ext_r = [u_ghost_axis_ext; u; u_ghost_wall_ext]; % (n_r+3, n_z)
30
31 % ---- interior slice i=2..n_r
   -----
32 u_int = u(2:end-1, :); % (n_r-1, n_z)
33
34 % --- Radial convective flux (1/r) d(r*u^2)/dr ---
35 u_r_im2 = u_ext_r(1:n_r-1, :);
36 u_r_im1 = u_ext_r(2:n_r, :);
37 u_r_i = u_ext_r(3:n_r+1, :); % = u_int
38 u_r_ip1 = u_ext_r(4:n_r+2, :);
39 u_r_ip2 = u_ext_r(5:n_r+3, :);
40
41 u_out = 0.5 * (u_int + u(3:end, :));
42 u_in = 0.5 * (u(1:end-2, :) + u_int);
43

```

```

44     u_out_upwind = upwind2_face(u_r_im1, u_r_i, u_r_ip1, u_r_ip2, u_out)
45 ;
46     u_in_upwind  = upwind2_face(u_r_im2, u_r_im1, u_r_i, u_r_ip1, u_in);
47
48     % ---- face velocities for axial convection
49     -----
50     w_at_rf = 0.5 * (w(1:end-1, :) + w(2:end, :));    % (n_r-1, n_z+1)
51     w_top   = w_at_rf(:, 2:end);
52     w_bot   = w_at_rf(:, 1:end-1);
53
54     % --- Axial convective flux d(u*w)/dz --- (2nd-order upwind,
55     interior rows)
56     u_z_jm2 = u_zext2(2:end-1, 1:n_z);
57     u_z_jm1 = u_zext2(2:end-1, 2:n_z+1);
58     u_z_j    = u_zext2(2:end-1, 3:n_z+2);    % = u_int
59     u_z_jp1  = u_zext2(2:end-1, 4:n_z+3);
60     u_z_jp2  = u_zext2(2:end-1, 5:n_z+4);
61
62     u_top_upwind = upwind2_face(u_z_jm1, u_z_j, u_z_jp1, u_z_jp2, w_top)
63 ;
64     u_bot_upwind = upwind2_face(u_z_jm2, u_z_jm1, u_z_j, u_z_jp1, w_bot)
65 ;
66
67     % ---- reshaping for broadcasting
68     -----
69     rc_o    = r_c(2:end);
70     rc_i    = r_c(1:end-1);
71     rf_int  = r_f(2:end-1);
72
73     % ---- convective fluxes (2nd-order upwind)
74     -----
75     conv_r = (rc_o .* u_out .* u_out_upwind - rc_i .* u_in .*
76     u_in_upwind) ./ (rf_int .* dr);
77     conv_z = (w_top .* u_top_upwind - w_bot .* u_bot_upwind) / dz;
78
79     % ---- diffusive fluxes (central, 2nd order)
80     -----
81     visc_r = ((rc_o .* (u(3:end, :) - u_int) - rc_i .* (u_int - u(1:end
82     -2, :))) ./ (rf_int * dr^2) ...
83     - u_int ./ rf_int.^2) / Re;
84
85     visc_z = (u_ext(2:end-1, 3:end) - 2*u_int + u_ext(2:end-1, 1:end-2))
86     / dz^2 / Re;
87
88     rhs = -conv_r - conv_z + visc_r + visc_z;
89 end

```

Listing 8: compute_rhs_u.m — Radial momentum RHS: 2nd-order upwind convection, central diffusion

```

1 function [u_new, w_new, p_new, T_new, rhs_u, rhs_w, rhs_T] = ...
2     ab2_step(u, w, p, T, rhs_u_old, rhs_w_old, rhs_T_old, ...
3     r_c, r_f, dr, dz, dt, Re, Pr, n_r, n_z, A_poisson, alpha_p)
4 % AB2_STEP Adams-Bashforth 2 with one incremental pressure correction.
5 % The first call uses forward Euler when the previous RHS arrays are
6 % empty.
7
8     rhs_u = compute_rhs_u(u, w, r_c, r_f, dr, dz, Re);

```

```

8   rhs_w = compute_rhs_w(w, u, r_c, r_f, dr, dz, Re);
9   rhs_T = compute_rhs_T(T, u, w, r_c, r_f, dr, dz, Re, Pr);
10
11   if isempty(rhs_u_old)
12       explicit_u = rhs_u;
13       explicit_w = rhs_w;
14       explicit_T = rhs_T;
15   else
16       explicit_u = 1.5 * rhs_u - 0.5 * rhs_u_old;
17       explicit_w = 1.5 * rhs_w - 0.5 * rhs_w_old;
18       explicit_T = 1.5 * rhs_T - 0.5 * rhs_T_old;
19   end
20
21   % The old pressure gradient makes the Poisson unknown a correction p
22   ,.
23   u_star = u;
24   grad_p_r = (p(2:end, :) - p(1:end-1, :)) / dr;
25   u_star(2:end-1, :) = u(2:end-1, :) ...
26       + dt * (explicit_u - grad_p_r);
27
28   w_star = w;
29   grad_p_z = (p(:, 2:end) - p(:, 1:end-1)) / dz;
30   w_star(:, 2:end-1) = w(:, 2:end-1) ...
31       + dt * (explicit_w - grad_p_z);
32   u_star = apply_bc_u(u_star);
33   w_star = apply_bc_w(w_star);
34
35   p_prime = solve_poisson(A_poisson, u_star, w_star, ...
36       r_c, r_f, dr, dz, dt, n_r, n_z);
37   [u_new, w_new] = correct_velocity(u_star, w_star, ...
38       p_prime, dr, dz, dt);
39   u_new = apply_bc_u(u_new);
40   w_new = apply_bc_w(w_new);
41
42   p_new = update_pressure(p, p_prime, alpha_p);
43   T_new = T + dt * explicit_T;
44 end

```

Listing 9: ab2_step.m — AB2 step with one incremental pressure correction

B.5 Poisson Solver and Pressure Correction

```

1 function A = build_poisson_matrix(r_c, r_f, dr, dz, n_r, n_z)
2 % BUILD_POISSON_MATRIX Assemble sparse Poisson matrix (called once).
3 %
4 %   A = build_poisson_matrix(r_c, r_f, dr, dz, n_r, n_z)
5 %
6 %   Returns A : sparse matrix (n_r*n_z x n_r*n_z)
7 %
8 %   Flat index: k = (i-1)*n_z + j    (1-based)
9 %
10 %   BCs for p':
11 %       Centreline (i=1):      Neumann (r_f(1)=0, flux vanishes naturally)
12 %       Wall (i=n_r):          Neumann (outer flux = 0)
13 %       Inlet (j=1):           Neumann (south flux = 0)
14 %       Outlet (j=n_z):        Dirichlet p' = 0
15

```

```

16 N = n_r * n_z;
17
18 % Pre-allocate triplets (at most 5 entries per row)
19 rows = zeros(5*N, 1);
20 cols = zeros(5*N, 1);
21 vals = zeros(5*N, 1);
22 idx = 0;
23
24 for i = 1:n_r
25     for j = 1:n_z
26         k = (i-1)*n_z + j;
27
28         % Dirichlet at outlet (j = n_z): p' = 0
29         if j == n_z
30             idx = idx + 1;
31             rows(idx) = k; cols(idx) = k; vals(idx) = 1.0;
32             continue;
33         end
34
35         % Coefficients from cylindrical Poisson
36         rf_o = r_f(i + 1); % r_{i+1/2}
37         rf_i_val = r_f(i); % r_{i-1/2}
38         rc = r_c(i);
39
40         coeff_E = rf_o / (rc * dr^2); % p'(i+1, j)
41         coeff_W = rf_i_val / (rc * dr^2); % p'(i-1, j)
42         coeff_N = 1.0 / dz^2; % p'(i, j+1)
43         coeff_S = 1.0 / dz^2; % p'(i, j-1)
44
45         % Neumann wall (i = n_r): outer flux = 0
46         if i == n_r
47             coeff_E = 0.0;
48         end
49
50         % Neumann inlet (j = 1): south flux = 0
51         if j == 1
52             coeff_S = 0.0;
53         end
54
55         % Centre coefficient
56         coeff_C = -(coeff_E + coeff_W + coeff_N + coeff_S);
57
58         % Diagonal
59         idx = idx + 1;
60         rows(idx) = k; cols(idx) = k; vals(idx) = coeff_C;
61
62         % East (i+1, j)
63         if i < n_r
64             idx = idx + 1;
65             rows(idx) = k; cols(idx) = i*n_z + j; vals(idx) =
coeff_E;
66         end
67
68         % West (i-1, j)
69         if i > 1
70             idx = idx + 1;
71             rows(idx) = k; cols(idx) = (i-2)*n_z + j; vals(idx) =
coeff_W;

```

```

72         end
73
74         % North (i, j+1)
75         if j < n_z
76             idx = idx + 1;
77             rows(idx) = k; cols(idx) = (i-1)*n_z + (j+1); vals(idx)
= coeff_N;
78         end
79
80         % South (i, j-1)
81         if j > 1
82             idx = idx + 1;
83             rows(idx) = k; cols(idx) = (i-1)*n_z + (j-1); vals(idx)
= coeff_S;
84         end
85     end
86 end
87
88 % Trim and build sparse matrix
89 rows = rows(1:idx);
90 cols = cols(1:idx);
91 vals = vals(1:idx);
92 A = sparse(rows, cols, vals, N, N);
93 end

```

Listing 10: build_poisson_matrix.m — Sparse Poisson matrix assembly

```

1 function p_prime = solve_poisson(A, u_star, w_star, r_c, r_f, dr, dz, dt
, n_r, n_z)
2 % SOLVE_POISSON Solve A * p' = b for pressure correction.
3 %
4 % p_prime = solve_poisson(A, u_star, w_star, r_c, r_f, dr, dz, dt, n_r
, n_z)
5 %
6 % RHS: b(i,j) = (1/dt) * div(u*)
7 % Returns p_prime : (n_r, n_z)
8
9 rf_o = r_f(2:end); % (n_r, 1)
10 rf_i = r_f(1:end-1); % (n_r, 1)
11 rc = r_c; % (n_r, 1)
12
13 % Radial divergence
14 div_u = (rf_o .* u_star(2:end, :) - rf_i .* u_star(1:end-1, :)) ./ (
rc * dr);
15
16 % Axial divergence
17 div_w = (w_star(:, 2:end) - w_star(:, 1:end-1)) / dz;
18
19 b = (div_u + div_w) / dt; % (n_r, n_z)
20
21 % Outlet row (j=n_z) has Dirichlet p'=0 -> RHS = 0
22 b(:, end) = 0.0;
23
24 % Flatten and solve
25 b_flat = b(:); % column-major flatten (same as reshape(
b, [], 1))
26 % MATLAB sparse direct solve (backslash)
27 % Note: MATLAB stores column-major, Python row-major. The flat index

```

```

28 % must match the matrix assembly order: k = (i-1)*n_z + j.
29 % Reshaping b column-major: b(:) goes column by column (j varies
30 fastest
31 % for fixed i when b is (n_r, n_z)). But our matrix uses k=(i-1)*n_z
32 +j,
33 % where j varies fastest for fixed i. This matches b(:) since MATLAB
34 % flattens column-major on (n_r, n_z) -> i varies fastest.
35 % We need to transpose to match: b' is (n_z, n_r), then b'(:) has
36 % j varying slowest... no. Let me be precise.
37 %
38 % Matrix assembly: k = (i-1)*n_z + j, so for i=1, j=1..n_z -> k=1..
39 n_z.
40 % MATLAB b(:) for b of size (n_r, n_z) gives b(1,1), b(2,1), ..., b(
41 n_r,1),
42 % b(1,2), ..., i.e. i varies fastest. But we need j to vary
43 fastest.
44 % Solution: transpose b before flattening.
45 b_flat = reshape(b', [], 1); % b' is (n_z, n_r), flatten -> j
46 varies fastest
47
48 p_prime_flat = A \ b_flat;
49
50 % Reshape back: p_prime_flat has k=(i-1)*n_z+j ordering
51 p_prime = reshape(p_prime_flat, n_z, n_r)'; % (n_r, n_z)
52 end

```

Listing 11: solve_poisson.m — Poisson solve using MATLAB backslash

```

1 function [u_new, w_new] = correct_velocity(u_star, w_star, p_prime, dr,
2 dz, dt)
3 % CORRECT_VELOCITY Apply pressure correction to obtain div-free
4 velocities.
5 %
6 % u^{n+1}(i,j) = u*(i,j) - dt * (p'(i,j) - p'(i-1,j)) / dr
7 % w^{n+1}(i,j) = w*(i,j) - dt * (p'(i,j) - p'(i,j-1)) / dz
8
9 u_new = u_star;
10 w_new = w_star;
11
12 % Radial velocity at interior r-faces (rows 2:end-1)
13 u_new(2:end-1, :) = u_star(2:end-1, :) ...
14 - dt * (p_prime(2:end, :) - p_prime(1:end-1, :)) / dr;
15
16 % Axial velocity at interior z-faces (columns 2:end-1)
17 w_new(:, 2:end-1) = w_star(:, 2:end-1) ...
18 - dt * (p_prime(:, 2:end) - p_prime(:, 1:end-1)) / dz;
19 end

```

Listing 12: correct_velocity.m — Velocity correction step

```

1 function p_new = update_pressure(p, p_prime, zeta)
2 % UPDATE_PRESSURE Accumulate relaxed correction: p_new = p + zeta*
3 p_prime.
4
5 p_new = p + zeta * p_prime;
6 end

```

Listing 13: update_pressure.m — Additive pressure-correction update

B.6 Temperature Solver

```

1 function rhs = compute_rhs_T(T, u, w, r_c, r_f, dr, dz, Re, Pr)
2 % COMPUTE_RHS_T Explicit RHS of the energy equation (no time
3 % discretization): rhs = -conv_r - conv_z + visc_r + visc_z
4 %
5 % Reused as the current and stored previous RHS in ab2_step.m.
6 %
7 % Convective fluxes use a 2nd-order upwind (linear-extrapolation / LUD
8 % )
9 % scheme -- see upwind2_face.m. Diffusive terms use standard 2nd-order
10 % central differencing.
11 %
12 % Wall BC: dtheta/dr* = +Re*Pr (see Report/nusselt_number_analysis.md
13 % for the derivation -- NOT -1, which drops the Re*Pr factor from the
14 % non-dimensionalization).
15 % Implemented as a ghost cell: theta_ghost(n_r+1,:) = theta(n_r,:) +
16 % dr*Re*Pr
17 %
18 % T : (n_r, n_z) current theta (BCs already applied)
19 % u : (n_r+1, n_z) radial velocity at r-faces
20 % w : (n_r, n_z+1) axial velocity at z-faces
21 % r_c, r_f, dr, dz, Re, Pr : as usual
22 %
23 % Returns rhs : (n_r, n_z)
24
25 [n_r, n_z] = size(T);
26 alpha = 1.0 / (Re * Pr);
27 RePr = Re * Pr;
28
29 % ---- ghost extension in r (depth 1, used by diffusion)
30 -----
31 T_ghost_axis = T(1, :);
32 T_ghost_wall = T(end, :) + dr * RePr;
33 T_ext = [T_ghost_axis; T; T_ghost_wall]; % (n_r+2, n_z)
34
35 % ---- ghost extension in z (depth 1, used by diffusion)
36 -----
37 T_ghost_in = -T(:, 1);
38 T_ghost_out = T(:, end);
39 T_zext = [T_ghost_in, T, T_ghost_out]; % (n_r, n_z+2)
40
41 % ---- second ghost layer, needed by the 2nd-order-upwind stencil
42 -----
43 T_ghost_axis2 = T(2, :);
44 T_ghost_wall2 = T(end, :) + 2.0 * dr * RePr;
45 T_ext2 = [T_ghost_axis2; T_ghost_axis; T; T_ghost_wall;
46 T_ghost_wall2]; % (n_r+4, n_z)
47
48 T_ghost_in2 = -T(:, 2);
49 T_ghost_out2 = T(:, end);
50 T_zext2 = [T_ghost_in2, T_ghost_in, T, T_ghost_out, T_ghost_out2];
51 % (n_r, n_z+4)
52
53 % ---- radial convective flux (1/r) d(r*u*theta)/dr
54 -----
55 u_E = u(2:end, :);
56 u_I = u(1:end-1, :);

```

```

49
50 T_r_im2 = T_ext2(1:n_r,      :);
51 T_r_im1 = T_ext2(2:n_r+1,    :);
52 T_r_i    = T_ext2(3:n_r+2,    :);    % = T
53 T_r_ip1  = T_ext2(4:n_r+3,    :);
54 T_r_ip2  = T_ext2(5:n_r+4,    :);
55
56 T_rf_o_upwind = upwind2_face(T_r_im1, T_r_i, T_r_ip1, T_r_ip2, u_E);
57 T_rf_i_upwind = upwind2_face(T_r_im2, T_r_im1, T_r_i, T_r_ip1, u_I);
58
59 rf_o = r_f(2:end);
60 rf_i = r_f(1:end-1);
61 rc   = r_c;
62
63 conv_r = (rf_o .* u_E .* T_rf_o_upwind - rf_i .* u_I .*
64 T_rf_i_upwind) ./ (rc .* dr);
65
66 % ---- axial convective flux d(w*theta)/dz
67 -----
68 w_tp = w(:, 2:end);
69 w_bt = w(:, 1:end-1);
70
71 T_z_jm2 = T_zext2(:, 1:n_z);
72 T_z_jm1 = T_zext2(:, 2:n_z+1);
73 T_z_j    = T_zext2(:, 3:n_z+2);    % = T
74 T_z_jp1  = T_zext2(:, 4:n_z+3);
75 T_z_jp2  = T_zext2(:, 5:n_z+4);
76
77 T_zf_p_upwind = upwind2_face(T_z_jm1, T_z_j, T_z_jp1, T_z_jp2, w_tp)
78 ;
79 T_zf_m_upwind = upwind2_face(T_z_jm2, T_z_jm1, T_z_j, T_z_jp1, w_bt)
80 ;
81
82 % At the physical inlet face the incoming transported value is the
83 % prescribed Dirichlet value theta=0, not a LUD reconstruction. The
84 % odd
85 % ghosts remain necessary for the second-order axial diffusion
86 % stencil.
87 inflow = w_bt(:, 1) > 0.0;
88 T_zf_m_upwind(inflow, 1) = 0.0;
89
90 conv_z = (w_tp .* T_zf_p_upwind - w_bt .* T_zf_m_upwind) / dz;
91
92 % ---- radial diffusion (1/r) d/dr(r*dtheta/dr)
93 -----
94 visc_r = alpha * (rf_o .* (T_ext(3:end, :) - T) ...
95                 - rf_i .* (T - T_ext(1:end-2, :))) ./ (rc * dr^2);
96
97 % ---- axial diffusion d^2 theta/dz^2
98 -----
99 visc_z = alpha * (T_zext(:, 3:end) - 2.0 * T + T_zext(:, 1:end-2)) /
100 dz^2;
101
102 rhs = -conv_r - conv_z + visc_r + visc_z;
103 end

```

Listing 14: compute_rhs_T.m — Energy equation RHS: 2nd-order upwind convection, central diffusion

B.7 Convergence Monitoring

```
1 function res = compute_residuals(u_new, u_old, w_new, w_old, T_new,
   T_old, r_c, r_f, dr, dz, dt)
2 % COMPUTE_RESIDUALS Compute convergence residuals.
3 %
4 % Returns res struct with fields:
5 % .continuity : max |div(u)|
6 % .momentum : max change rate of both u and w (interior)
7 % .temperature : max |T_new - T_old| / dt
8
9 div = compute_divergence(u_new, w_new, r_c, r_f, dr, dz);
10 res.continuity = max(abs(div(:)));
11 R_u = max(abs(u_new(2:end-1, :) - u_old(2:end-1, :)), [], 'all');
12 R_w = max(abs(w_new(:, 2:end-1) - w_old(:, 2:end-1)), [], 'all');
13 res.momentum = max(R_u, R_w) / dt;
14 res.temperature = max(abs(T_new(:) - T_old(:))) / dt;
15 end
```

Listing 15: compute_residuals.m — Residual computation

```
1 function converged = check_convergence(res, tol_cont, tol_vel, tol_temp)
2 % CHECK_CONVERGENCE Return true when all residuals are below tolerances
3 .
4 converged = (res.continuity < tol_cont) && ...
5             (res.momentum < tol_vel) && ...
6             (res.temperature < tol_temp);
7 end
```

Listing 16: check_convergence.m — Convergence check

References

- [1] R. K. Shah and A. L. London, *Laminar Flow Forced Convection in Ducts*, Supplement 1 to *Advances in Heat Transfer*, Academic Press, New York, 1978. (Circular duct, H1 boundary condition: $Nu = 48/11$.)
- [2] F. P. Incropera and D. P. DeWitt, *Fundamentals of Heat and Mass Transfer*, Wiley, New York. (Internal flow chapter: fully developed laminar flow in a circular tube with uniform surface heat flux, $Nu_D = 48/11 \approx 4.36$.)
- [3] W. M. Kays, M. E. Crawford, and B. Weigand, *Convective Heat and Mass Transfer*, McGraw-Hill, New York. (Fully developed laminar circular-tube flow, constant heat rate per unit length: $Nu = 48/11$.)